Extended Review

Worked answer key: question, given, diagram, steps with reasons, final answer

Conventions used throughout

• **Notation:** v_1 = initial velocity, v_2 = final velocity, Δd = displacement, Δt = time interval, a = acceleration, $\vec{a}_{av}$ = average acceleration, v_{ave} = average speed. An arrow marks a vector; bars, as in $|\Delta \vec{v}|$, mean its size only. In relative-velocity problems v_{AB} means the velocity of A relative to B.

• **Which object?** When two objects move in one problem, a word in the subscript says which one a symbol belongs to. For example, a_{car} means the car's acceleration.

• **Constant-acceleration equations** (valid only when a is constant): $v_2 = v_1 + a\Delta t$; $\Delta d = \frac{1}{2}(v_1 + v_2)\Delta t$; $\Delta d = v_1\Delta t + \frac{1}{2}a\Delta t^2$; $\Delta d = v_2\Delta t - \frac{1}{2}a\Delta t^2$; $v_2^2 = v_1^2 + 2a\Delta d$.

• **Signs:** each question states which direction is positive. A negative value means the opposite direction. Final answers state the real direction.

• **$g = 9.81\ m/s^2$** and no air resistance. Full precision is kept in every step; only final answers are rounded.

REVIEW 1

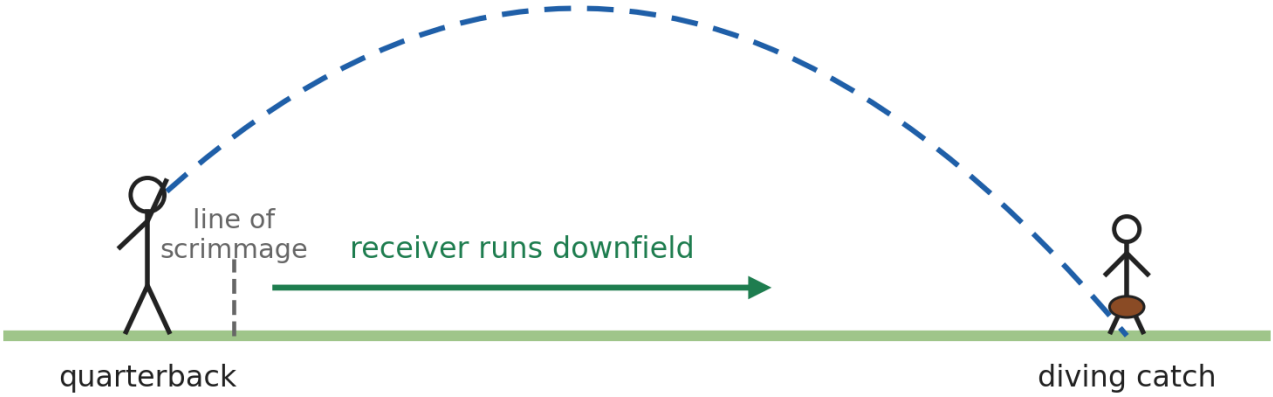
Football pass and receiver

A football quarterback attempts a pass to one of the receivers. As the ball is snapped, the receiver leaves the line of scrimmage and runs directly down field. The quarterback releases the ball 2.0 s later and from a position 3.0 m behind the line of scrimmage. He throws the ball with a speed of 26 m/s at an elevation of 60° above the horizontal. The receiver makes a diving reception, catching the ball just as it reaches the ground. See the diagram below.

a. What is the time of flight of the football?

b. What is the average speed of the receiver?

Picture the situation



At the snap the receiver starts running downfield. Two seconds later the quarterback, standing 3.0 m behind the line, throws a high pass. The receiver dives and catches it just as it reaches the ground.

SOLVING FOR

(a) how long the ball is in the air, Δt; (b) the receiver's average speed from the snap to the catch.

GIVEN

$v_1 = 26$ m/s at 60°	$\Delta d_y = -2.0$ m	$a_y = -9.81$ m/s^2
$x_{release} = -3.0$ m	delay $= 2.0$ s	$\Delta t = ?$

Sign convention: +x is downfield and +y is up. The line of scrimmage is x = 0. The ball ends 2.0 m lower than it starts (released 2.0 m up, caught at the ground), so $\Delta d_y = -2.0$ m.

Diagram: the play (drawn to scale)

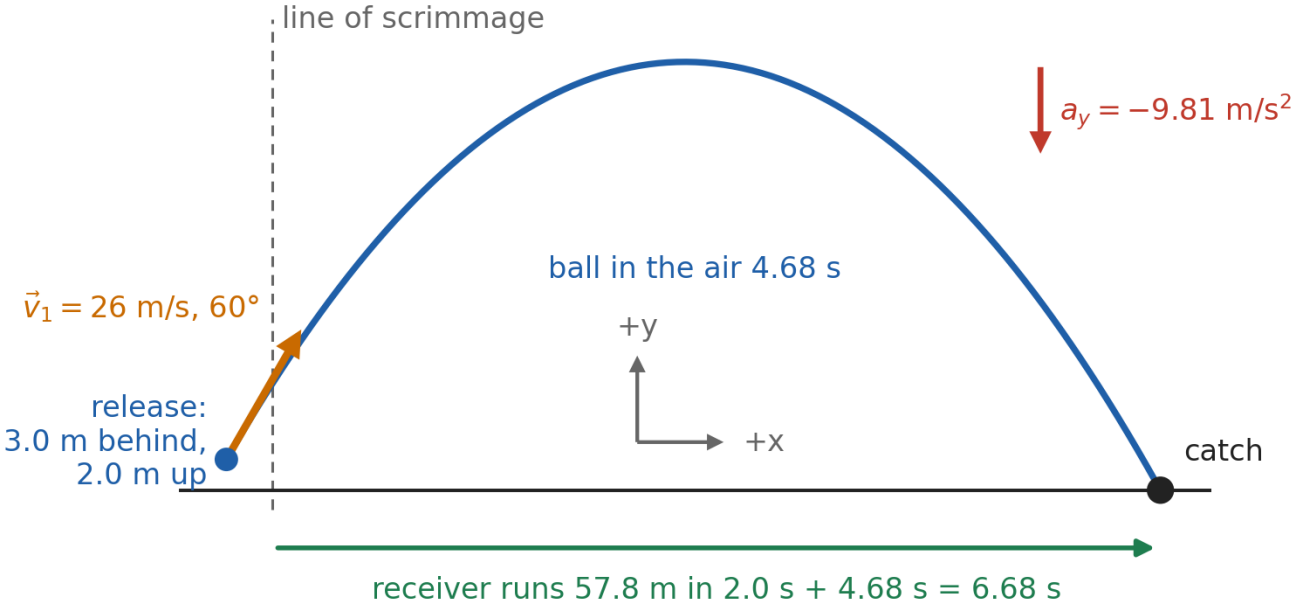
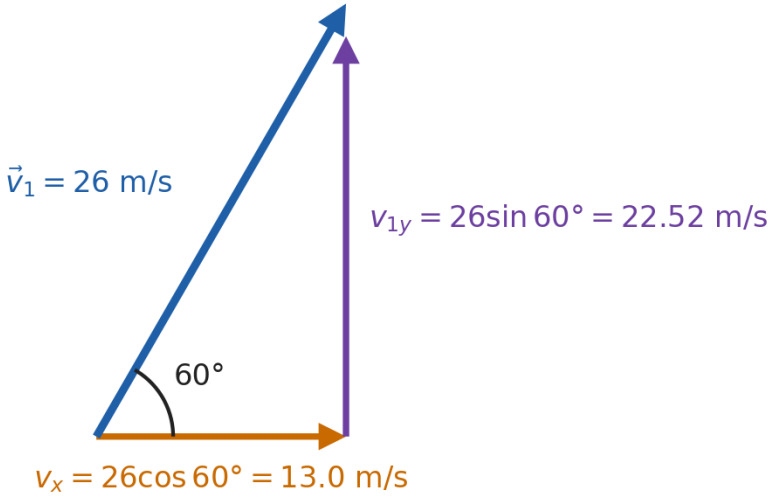


Diagram: splitting the launch velocity into parts



Step 1 – (a) Time in the air, from the vertical motion

This is a projectile, so we treat up-down and sideways separately. The flight ends when the ball reaches the ground, which is a fact about its height, so the vertical motion gives the time. Vertically we know the up-down part of the launch velocity, v_{1y}, plus a_y and Δd_y. Putting them into the displacement equation gives a quadratic in Δt.

Formula: $v_{1y} = v_1 \sin\theta = 26 \sin 60° = 22.52$ m/s

Formula: $\Delta d_y = v_{1y}\Delta t + \frac{1}{2} a_y \Delta t^2$

Put in the numbers:

$$-2.0 = 22.52\,\Delta t + \frac{1}{2}(-9.81)\,\Delta t^2 = 22.52\,\Delta t - 4.905\,\Delta t^2$$

Move everything to the left side:

$$4.905\,\Delta t^2 - 22.52\,\Delta t - 2.0 = 0$$

Formula: $\Delta t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

Here $a = 4.905,\ b = -22.52,\ c = -2.0$:

$$\Delta t = \frac{22.52 \pm \sqrt{(-22.52)^2 - 4(4.905)(-2.0)}}{2(4.905)}$$

$$\Delta t = \frac{22.52 \pm \sqrt{507.2 + 39.2}}{9.81}$$

$$\Delta t = \frac{22.52 \pm 23.37}{9.81} \Rightarrow \Delta t = \frac{45.89}{9.81} = 4.68 \text{ s} \quad \text{or} \quad \frac{-0.85}{9.81} = -0.09 \text{ s}$$

The “−” choice gives a negative time (before the throw), so we reject it. **Δt = 4.68 s**.

Step 2 – Where the ball lands

Sideways the ball moves at a steady $v_x = 26 \cos 60° = 13.0$ m/s for those 4.68 s. That tells us how far it travels from the quarterback. The quarterback was 3.0 m behind the line, so we subtract 3.0 m to get the distance from the line, which is where the receiver started.

Formula: $v_x = v_1 \cos\theta = 26 \cos 60° = 13.0 \text{ m/s}$

Formula: $\Delta d_x = v_x \Delta t$

$\Delta d_x = (13.0 \text{ m/s})(4.678 \text{ s}) = 60.81 \text{ m}$

$\text{catch point} = 60.81 - 3.0 = 57.81 \text{ m past the line}$

Step 3 – (b) The receiver's average speed

The receiver ran from the line (at the snap) to the catch point. He ran during the 2.0 s before the throw *and* the 4.68 s the ball was in the air. $\Delta t_{receiver}$ is his total running time and d is the distance he ran. Average speed = total distance ÷ total time.

$\Delta t_{receiver} = 2.0 + 4.678 = 6.678 \text{ s}$

Formula: $v_{ave} = \dfrac{d}{\Delta t}$

$v_{ave} = \dfrac{d}{\Delta t_{receiver}} = \dfrac{57.81 \text{ m}}{6.678 \text{ s}} = 8.66 \text{ m/s}$

(a) 4.7 s

(b) 8.7 m/s

REVIEW 2

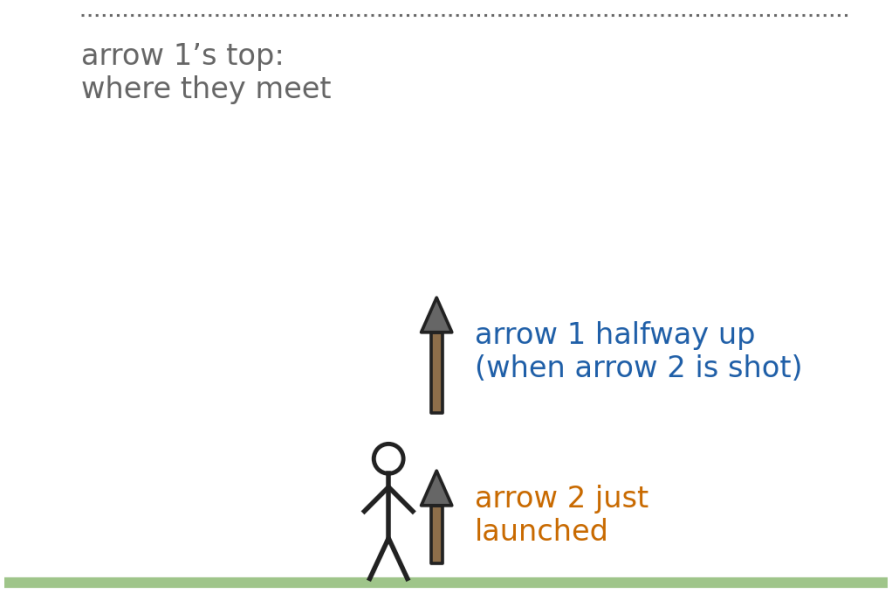
Two vertical arrows

An arrow is shot vertically upward with an initial speed of 25 m/s. When it's exactly halfway to the top of its flight, a second arrow is launched vertically upward from the same spot. The second arrow reaches the first arrow just as the first arrow reaches its highest point.

a. What is the launch speed of the second arrow?

b. What maximum height does the second arrow reach?

Picture the situation



The first arrow is already halfway up when the second one is fired from the same spot. The second arrow is faster and catches up with the first exactly at the first arrow's highest point.

GIVEN

$v_{1(\text{arrow 1})} = +25$ m/s

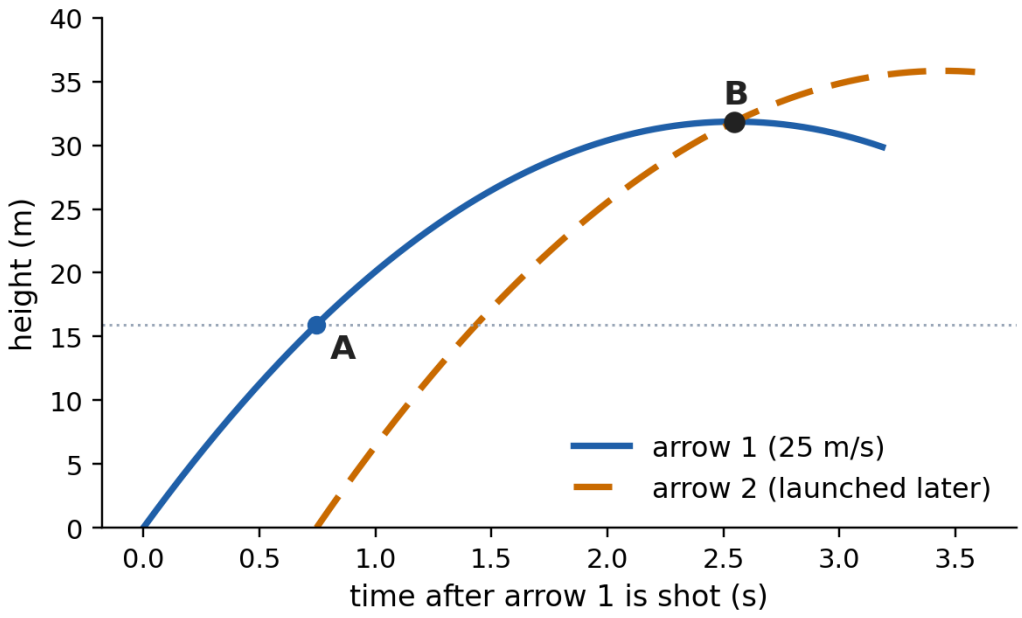
$a = -9.81$ m/s^2

$v_{1(\text{arrow 2})} = ?$

$H = ?$ (height of arrow 1's top)

Sign convention: up is positive, so a = −9.81 m/s^2 for both arrows. Each arrow has its own launch speed, so the label in brackets says which arrow a v_1 belongs to: $v_{1(\text{arrow 1})}$ is arrow 1's launch speed and $v_{1(\text{arrow 2})}$ is arrow 2's.

Graph: height vs time for both arrows (A: arrow 2 fired when arrow 1 is halfway up; B: they meet at the top)



Step 1 – Where and when arrow 1 reaches its top

Everything in this problem is measured against arrow 1's top, so find that first. At the top an arrow stops for an instant, so $v_2 = 0$. Use $v_2^2 = v_1^2 + 2a\Delta d$ for the height and $v_2 = v_1 + a\Delta t$ for the time. Here $v_1 = 25$ m/s (arrow 1's launch speed), and the displacement Δd up to the top is the height H we want. Δt_{top} is the time arrow 1 takes to reach its top.

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Put in $v_2 = 0$ and $\Delta d = H$: $0 = 25^2 + 2(-9.81)H = 625 - 19.62H$

Add $19.62H$ to both sides, then divide by 19.62:

$$H = \frac{625}{19.62} = 31.86 \text{ m}$$

Formula: $v_2 = v_1 + a\Delta t$

Put in $v_2 = 0$: $0 = 25 + (-9.81)\Delta t_{top}$

Add $9.81\Delta t_{top}$ to both sides, then divide by 9.81:

$$\Delta t_{top} = \frac{25}{9.81} = 2.548 \text{ s}$$

Step 2 – When arrow 1 is halfway up

Arrow 2 is fired when arrow 1 is at H/2 = 15.93 m (on the way up). We need that moment. First find arrow 1's speed at that height (no time needed), then use the speed to get the time. Here v_2 is arrow 1's speed at the halfway point, and Δt_{half} is the time arrow 1 takes to get there.

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

$v_2^2 = 25^2 + 2(-9.81)(15.93) = 625 - 312.5 = 312.5$

Square root both sides: $v_2 = \sqrt{312.5} = 17.68$ m/s

Formula: $v_2 = v_1 + a\Delta t$

$17.68 = 25 + (-9.81)\Delta t_{half}$

Subtract 25 from both sides: $-7.32 = -9.81\,\Delta t_{half}$

Divide both sides by -9.81: $\Delta t_{half} = \dfrac{-7.32}{-9.81} = 0.746$ s

Step 3 – (a) Launch speed of arrow 2

Arrow 2 is fired at 0.746 s and must be at 31.86 m when arrow 1 gets there, at 2.548 s. So arrow 2 flies for 2.548 − 0.746 = 1.802 s and rises 31.86 m in that time. In $\Delta d = v_1\Delta t + \frac{1}{2}a\Delta t^2$, v_1 is the starting velocity of whatever is moving. The thing moving here is arrow 2, so its launch speed $v_{1(\text{arrow 2})}$ goes in the v_1 spot. It is the only unknown:

Formula: $\Delta d = v_1\Delta t + \dfrac{1}{2}a\Delta t^2$

$31.86 = v_{1(\text{arrow 2})}(1.802) + \dfrac{1}{2}(-9.81)(1.802)^2$

Work out the last term: $31.86 = 1.802\,v_{1(\text{arrow 2})} - 15.93$

Add 15.93 to both sides: $47.79 = 1.802\,v_{1(\text{arrow 2})}$

Divide both sides by 1.802: $v_{1(\text{arrow 2})} = \dfrac{47.79}{1.802} = 26.52$ m/s

Step 4 – (b) How high arrow 2 goes

When they meet, arrow 2 is still moving up (it was launched faster), so it keeps rising after passing arrow 1. Its own top is where its velocity reaches zero. This time the motion is arrow 2's, so v_1 is $v_{1(\text{arrow 2})} = 26.52$ m/s, and Δd is arrow 2's maximum height, $H_{\text{arrow 2}}$:

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Put in $v_2 = 0$ and $\Delta d = H_{\text{arrow 2}}$:

$$0 = (26.517)^2 + 2(-9.81)H_{\text{arrow 2}} = 703.1 - 19.62H_{\text{arrow 2}}$$

Add $19.62H_{\text{arrow 2}}$, then divide by 19.62:

$$H_{\text{arrow 2}} = \frac{703.1}{19.62} = 35.84 \text{ m}$$

(a) 27 m/s [up]

(b) 36 m above the launch point

Check: arrow 2's speed when they meet is 26.52 − 9.81(1.802) = 8.84 m/s [up]. It is still positive, so arrow 2 really is still rising and does go higher than 31.9 m.

REVIEW 3

Truck and accelerating car

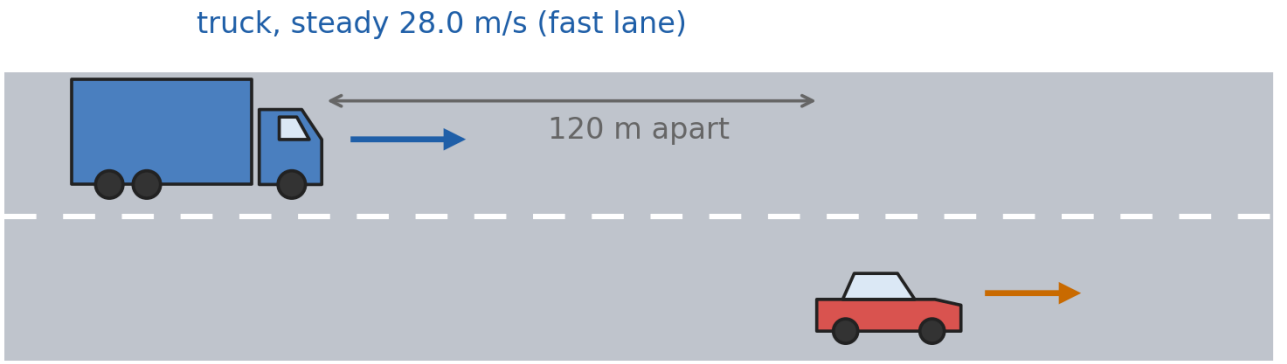
A truck travels at a constant speed of 28.0 m/s in the fast lane of a two-lane highway. It approaches a stationary car stopped at the side of the road. When the truck is still 1.2 x 10^2 m behind the car, the car pulls out into the slow lane with an acceleration of 2.6 m/s^2.

a. How long will it take the truck to pass the car?

b. How far will the car have travelled when the truck passes it?

c. If the car were to maintain this acceleration, what would be its speed when it overtakes the truck?

Picture the situation



The truck comes up from behind at a steady speed. The car starts from rest and speeds up. At first the truck is faster and passes the car; later the car is faster and passes the truck back.

GIVEN

$v_{truck} = 28.0$ m/s $\qquad a_{car} = 2.6$ m/s^2 $\qquad$ gap $= 1.2 \times 10^2$ m

Sign convention: + is the direction of travel. All positions are measured from where the car starts (x = 0), so the truck starts at x = −120 m.

Step 1 – Write each vehicle's position

"Passing" means both are at the same position at the same time. So we write each one's position as a formula in Δt, starting from its own starting position. In these formulas x is the position at time Δt and x_1 is the starting position. The truck moves at constant velocity; the car starts from rest with constant acceleration.

Formula (truck, constant velocity): $x = x_1 + v\,\Delta t$

$$x_{truck} = -120 + 28.0\,\Delta t$$

Formula (car): $x = x_1 + v_1\Delta t + \frac{1}{2}a\Delta t^2$

$$x_{car} = 0 + (0)\Delta t + \frac{1}{2}(2.6)\,\Delta t^2 = 1.3\,\Delta t^2$$

Step 2 – Set the positions equal

Setting $x_{car} = x_{truck}$ and moving everything to one side gives a quadratic.

$$1.3\,\Delta t^2 = -120 + 28.0\,\Delta t$$

Move everything to the left (add 120, subtract $28.0\Delta t$):

$$1.3\,\Delta t^2 - 28.0\,\Delta t + 120 = 0$$

Formula: $\Delta t = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

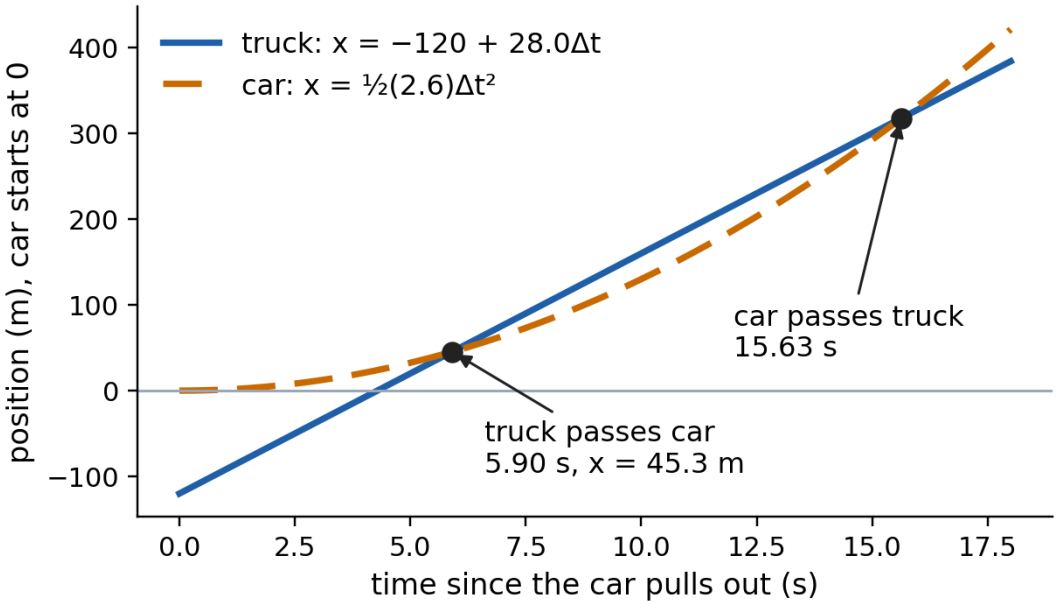
Here $a = 1.3,\ b = -28.0,\ c = 120$:

$$\Delta t = \frac{28.0 \pm \sqrt{784 - 624}}{2.6} = \frac{28.0 \pm \sqrt{160}}{2.6} = \frac{28.0 \pm 12.65}{2.6}$$

$$\Delta t = 5.90 \text{ s} \quad \text{or} \quad \Delta t = 15.63 \text{ s}$$

Both answers are real meetings. At 5.90 s the truck catches and passes the car. The car keeps speeding up, so at 15.63 s it catches and passes the truck. Part (a) asks about the first; part (c) about the second.

Graph: positions of the truck and the car



Step 3 – (b) How far the car has gone when the truck passes

Put the first time into the car's position formula (the car started at x = 0, so its position is its distance travelled).

$$x_{car} = 1.3\,\Delta t^2 = 1.3(5.904)^2 = 1.3(34.86) = 45.3 \text{ m}$$

Step 4 – (c) The car's speed when it overtakes the truck

Use the later time in $v_2 = v_1 + a\Delta t$ for the car: $v_1 = 0$ because the car started from rest, $a = a_{car} = 2.6$ m/s², and v_2 is the car's speed when it overtakes.

Formula: $v_2 = v_1 + a\Delta t$

$$v_2 = 0 + (2.6)(15.63) = 40.6 \text{ m/s}$$

(a) 5.9 s

(b) 45 m

(c) 41 m/s

Check: at 5.904 s the truck is at −120 + 28.0(5.904) = 45.3 m, the same place as the car.

REVIEW 4

Clock hand: speed and changing velocity

A clock has a second hand 15 cm long, from centre to tip. (Hint: the velocity of the tip of the second hand is always tangent to the circle)

a. Draw a labelled diagram to represent this situation

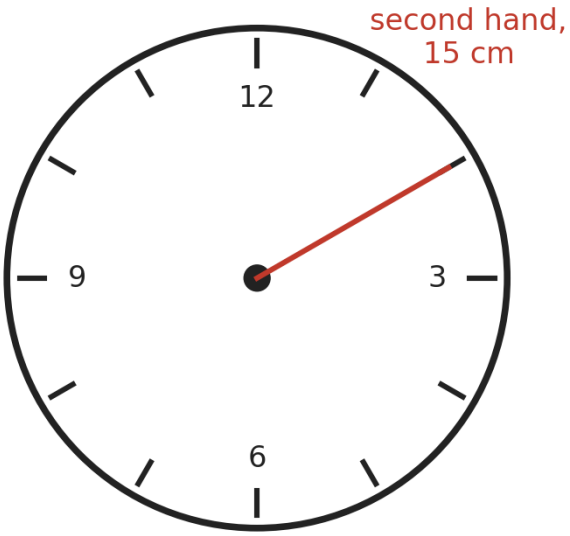
b. Calculate the instantaneous speed of the tip of the second hand.

c. What is the instantaneous velocity of the tip at 15 s?

d. What is its change in velocity between 30 s and 45 s?

e. What is its average acceleration during the same time interval? (1 mark)

Picture the situation



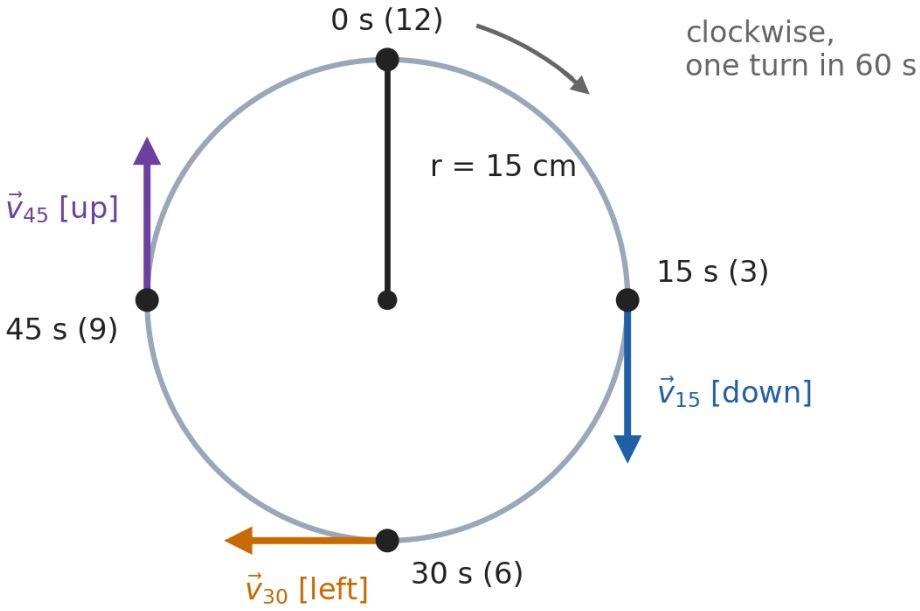
The tip of the second hand goes around a circle of radius 15 cm once every minute. Its speed never changes, but its direction keeps turning.

GIVEN

$r = 15$ cm $\qquad T = 60$ s per turn $\qquad \Delta t = 45 - 30 = 15$ s

Sign convention: up and right are positive. The hand turns clockwise and points at 12 at 0 s. r is the length of the hand (the radius of the circle) and T is the time for one full turn. A number in a velocity's subscript is the clock time: $\vec{v}_{15}$, $\vec{v}_{30}$ and $\vec{v}_{45}$ are the tip's velocity at 15 s, 30 s and 45 s.

Diagram (a): the clock face with velocities at 15 s, 30 s and 45 s



(b) Speed of the tip

In one full turn the tip travels once around the circle, a distance of 2πr (the circumference), and that takes one period, T = 60 s. Speed = distance ÷ time.

Formula: $v = \frac{d}{T}$ with $d = 2\pi r$ (circumference), $T = 60$ s

$$v = \frac{2\pi(15\text{ cm})}{60\text{ s}} = \frac{94.25\text{ cm}}{60\text{ s}} = 1.57\text{ cm/s}$$

(c) Velocity at 15 s

After 15 s the hand points at the 3. The hint says the velocity is tangent to the circle, along the direction of motion. Going clockwise from the 3, the tip is moving straight down.

$\vec{v}_{15}$ = 1.57 cm/s [down].

(d) Change in velocity from 30 s to 45 s

At 30 s the tip is at the 6 and moving left. At 45 s it is at the 9 and moving up. Same speed, different directions. $\Delta\vec{v} = \vec{v}_{45} + (-\vec{v}_{30})$, and reversing the leftward $\vec{v}_{30}$ gives an arrow pointing right. Up and right are perpendicular and equal in length, so Pythagoras gives the size and the angle is 45°.

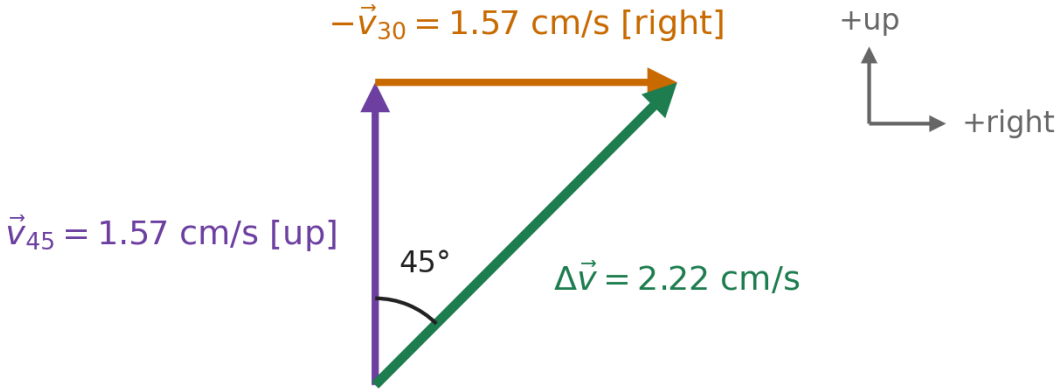
Formula: $\Delta\vec{v} = \vec{v}_{45} - \vec{v}_{30} = \vec{v}_{45} + (-\vec{v}_{30})$

Formula (Pythagoras): $c = \sqrt{a^2 + b^2}$

$$|\Delta\vec{v}| = \sqrt{(1.571)^2 + (1.571)^2} = \sqrt{4.936} = 2.22 \text{ cm/s}$$

The direction is 45° between up and right, which points toward the centre of the clock from where the tip is halfway through this interval (between the 6 and the 9).

Diagram (d): $\vec{v}_{45} + (-\vec{v}_{30})$, tip to tail



(e) Average acceleration

Divide the change in velocity by the 15 s it took. The direction stays the same. $\vec{a}_{av}$ is the average acceleration, and $|\Delta\vec{v}|$ (with bars) means the size of $\Delta\vec{v}$ only.

Formula: $\vec{a}_{av} = \dfrac{\Delta\vec{v}}{\Delta t}$

$\vec{a}_{av} = \dfrac{2.22 \text{ cm/s}}{15 \text{ s}} = 0.148 \text{ cm/s}^2$, same direction as $\Delta\vec{v}$

(b) 1.6 cm/s

(c) 1.6 cm/s [down]

(d) 2.2 cm/s [up 45° right] (up and to the right at 45°)

(e) 0.15 cm/s^2 [up 45° right]

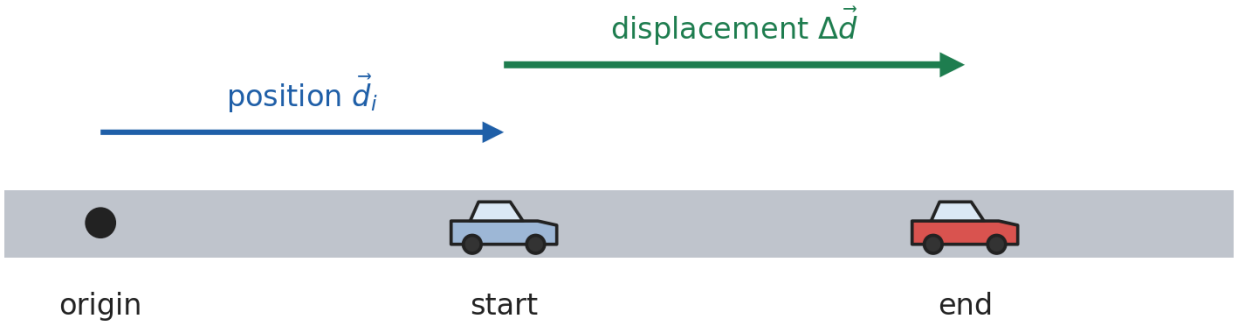
The acceleration points toward the inside of the circle. Any object moving in a circle at constant speed accelerates toward the centre, because its velocity is always being turned toward the centre.

CONCEPTS 1–2

Counts, velocity and displacement notation

1. How many people are there in this room? _____ How many significant digits is this? _____
2. Explain the difference between the following symbols:
   a. $\vec{v}_{av}$, $\Delta\vec{v}$, $\vec{v}_i$
   b. $\vec{d}$, $\Delta\vec{d}$

Picture the situation



Position is measured from a chosen origin. Displacement is the change in position: the arrow from where something started to where it ended.

1. Counting people

I do not know how many people were in your room, so count them. The key idea is that a count of whole objects is **exact**: there is no measuring error in "24 people". Exact numbers have unlimited (infinitely many) significant digits, so they never limit the significant figures of an answer.

2(a) $\vec{v}_{av}$, $\Delta\vec{v}$ and $\vec{v}_i$

All three have units of m/s, but they mean different things.

$$\vec{v}_{av} = \frac{\Delta\vec{d}}{\Delta t} \qquad \Delta\vec{v} = \vec{v}_f - \vec{v}_i$$

$\vec{\mathbf{v}}_{\mathbf{av}}$, average velocity: the displacement divided by the time taken (one velocity that describes the whole trip). $\boldsymbol{\Delta\vec{v}}$, change in velocity: how much the velocity changed, final minus initial. $\vec{\mathbf{v}}_{\mathbf{i}}$, initial velocity: the velocity at the start of the time interval ($\vec{v}_f$ is the velocity at the end).

2(b) $\vec{d}$ and $\Delta\vec{d}$

$$\Delta\vec{d} = \vec{d}_f - \vec{d}_i$$

$\vec{d}$, position: where an object is, measured from a chosen origin (as in the sketch). **$\Delta\vec{d}$**, displacement: the change in position, from the starting point to the ending point ($\vec{d}_i$ = initial position, $\vec{d}_f$ = final position). If you move the origin, every position changes, but the displacement between two points does not.

1. Your own head count; it is exact, so unlimited significant digits

2(a) average velocity; change in velocity; initial velocity

2(b) position (from an origin); displacement (change in position)

CONCEPT 3

Toddlers with different starting times

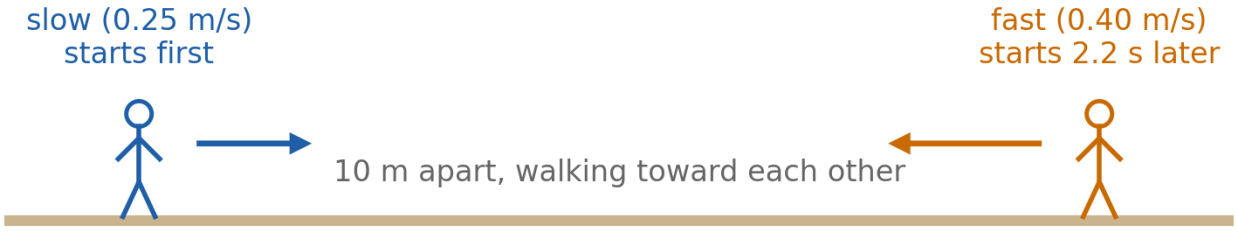
Two toddlers are 10 m apart and are moving towards each other, one travelling at 0.4 m/s and the other at 0.25 m/s. The faster child started 2.2 s after the slower one. Which of the following statements are true at the time the two children meet? (s = slow; f = fast)

a. $\Delta t_f = \Delta t_s + 2.2s$

b. $\Delta t_s = \Delta t_f + 2.2s$.

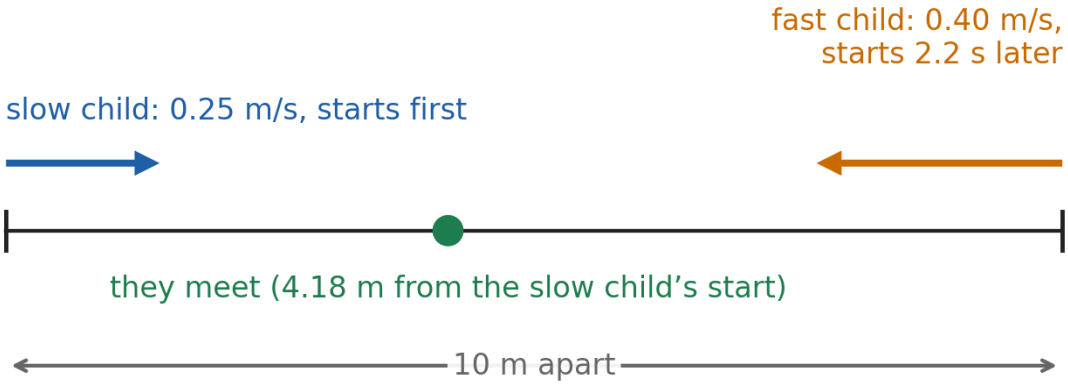
Explain your answer:

Picture the situation



Two toddlers walk toward each other. The slow one sets off first; 2.2 s later the fast one sets off.

Diagram: the situation (drawn to scale)



Explanation

Think about each toddler’s stopwatch. The slow toddler starts walking first. By the time the fast toddler starts, the slow toddler’s stopwatch already reads 2.2 s. They both stop when they meet. So the slow toddler has always been walking 2.2 s longer than the fast one.

$$\Delta t_s = \Delta t_f + 2.2 \text{ s}$$

Statement (a) says the opposite (the fast toddler walks longer), so it is false.

Optional: when and where they meet

The two walking distances must add up to the 10 m gap. d_s and d_f are the distances walked by the slow and fast toddler. Writing the slow toddler's time as $\Delta t_f + 2.2$ leaves one unknown.

Formula: $d = v\,\Delta t$

The distances fill the gap: $d_s + d_f = 10$

Put in $d_s = 0.25\,\Delta t_s = 0.25(\Delta t_f + 2.2)$ and $d_f = 0.40\,\Delta t_f$:

$0.25(\Delta t_f + 2.2) + 0.40\,\Delta t_f = 10$

Expand the bracket: $0.25\,\Delta t_f + 0.55 + 0.40\,\Delta t_f = 10$

Add the like terms: $0.65\,\Delta t_f + 0.55 = 10$

Subtract 0.55, then divide by 0.65: $\Delta t_f = \dfrac{9.45}{0.65} = 14.54$ s

$\Delta t_s = 14.54 + 2.2 = 16.74$ s

The slow toddler walks 0.25 × 16.74 = 4.18 m and the fast one 0.40 × 14.54 = 5.82 m (total 10 m).

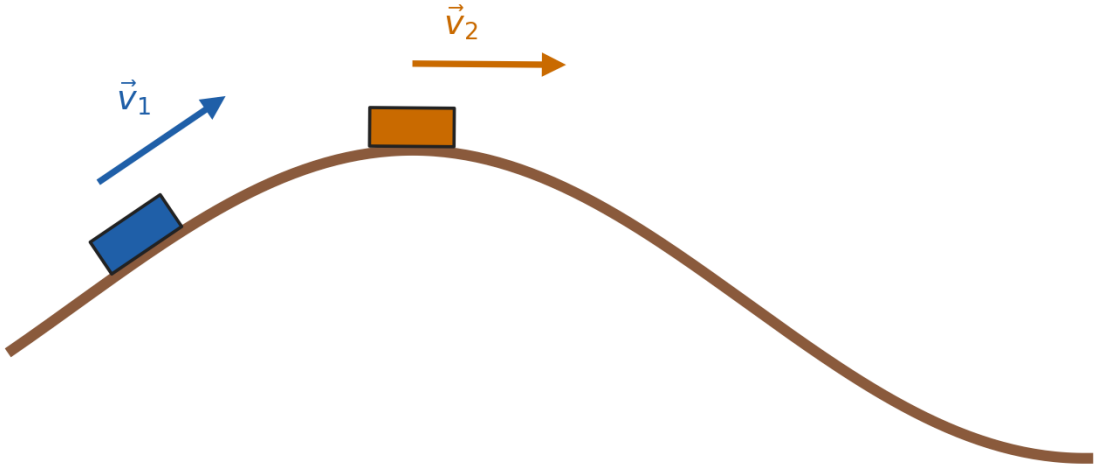
Only (b) is true: $\Delta t_s = \Delta t_f + 2.2$ s, because the slow child walked for 2.2 s longer.

CONCEPT 4

Roller coaster: acceleration direction

The following vectors show the velocity of a roller coaster at two different points in time. What is the direction of the acceleration? Explain.

Picture the situation



The coaster car climbs a hill (moving up and to the right), then levels off at the top (moving straight right).

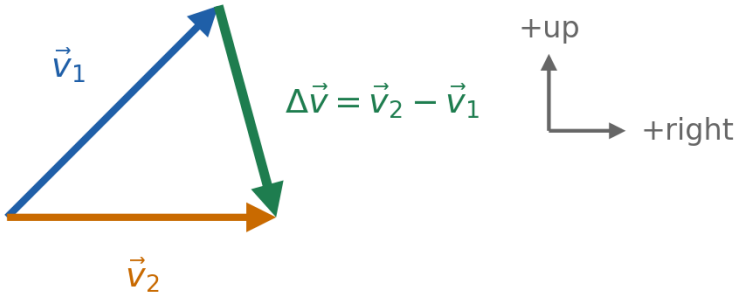
Step 1 – Acceleration points along the change in velocity

Average acceleration is the change in velocity divided by the time. Time is positive, so the acceleration points the same way as the change in velocity. So the question is really: which way did the velocity change?

$$\vec{a}_{av} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_2 - \vec{v}_1}{\Delta t}$$

Here $\vec{v}_1$ is the first velocity (climbing) and $\vec{v}_2$ is the later one (level at the top).

Diagram: tails together, $\Delta\vec{v}$ from the tip of $\vec{v}_1$ to the tip of $\vec{v}_2$



Step 2 – Read the direction off the diagram

Draw both velocities from the same starting point. The change $\Delta\vec{v}$ is the arrow from the tip of $\vec{v}_1$ to the tip of $\vec{v}_2$, since $\vec{v}_1 + \Delta\vec{v} = \vec{v}_2$. Here the upward part of the velocity disappears (a change downward) and the rightward part grows (a change to the right). So $\Delta\vec{v}$ points down and to the right.

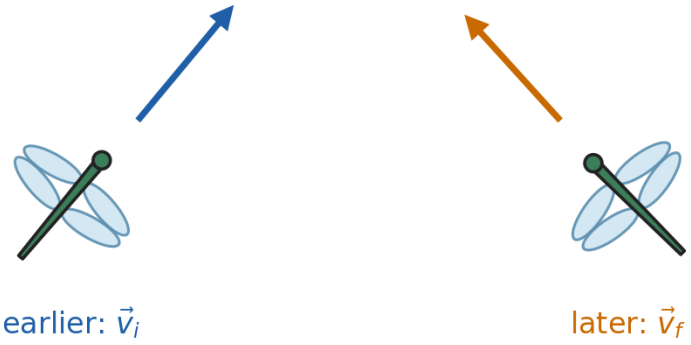
The acceleration points down and to the right, the direction of $\Delta\vec{v} = \vec{v}_2 - \vec{v}_1$.

CONCEPT 5

Dragonfly: subtract velocity vectors

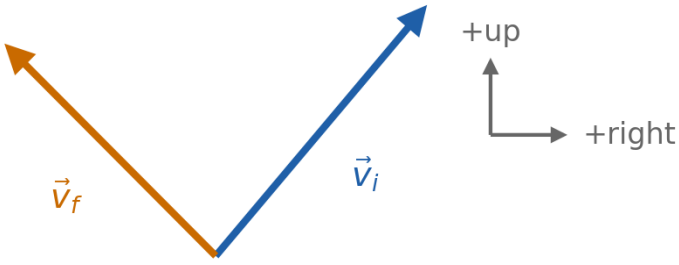
The velocity of a dragonfly at two different times is shown by the vectors below. Draw a vector diagram that would allow you to calculate the dragonfly's change in velocity.

Picture the situation



The dragonfly is first flying up and to the right, and later up and to the left.

The two velocities

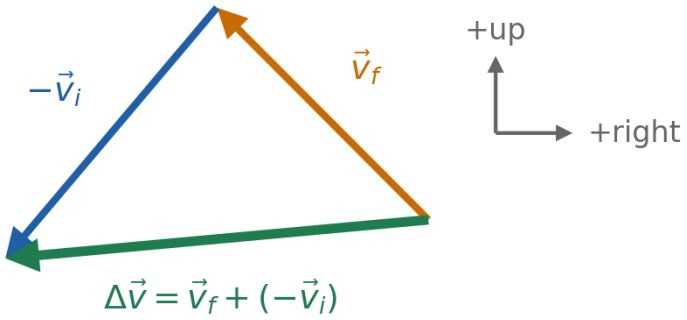


Step 1 – Turn the subtraction into addition

Change in velocity is final minus initial. We can only add arrows tip to tail, so we rewrite the subtraction as adding the opposite: reverse $\vec{v}_i$ (same length, opposite direction). Here $\vec{v}_i$ (i for initial) is the earlier velocity, up and to the right, and $\vec{v}_f$ (f for final) is the later one, up and to the left.

$$\Delta\vec{v} = \vec{v}_f - \vec{v}_i = \vec{v}_f + (-\vec{v}_i)$$

Vector diagram: $\vec{v}_f + (-\vec{v}_i)$, tip to tail



Step 2 – How to use the diagram

Draw $\vec{v}_f$. Put the tail of $-\vec{v}_i$ on its tip. The change in velocity, $\Delta\vec{v}$, is the arrow from the very first tail to the very last tip. To get numbers, measure $\Delta\vec{v}$ with a ruler and protractor using the worksheet's scale, or solve the triangle with the cosine law and sine law.

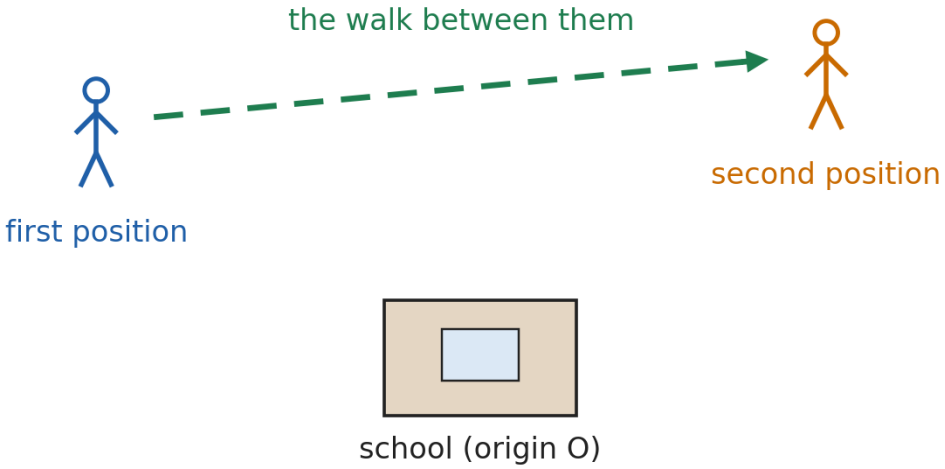
$\Delta\vec{v} = \vec{v}_f + (-\vec{v}_i)$, drawn tip to tail (it points left and slightly down)

CONCEPT 6

Student: subtract position vectors

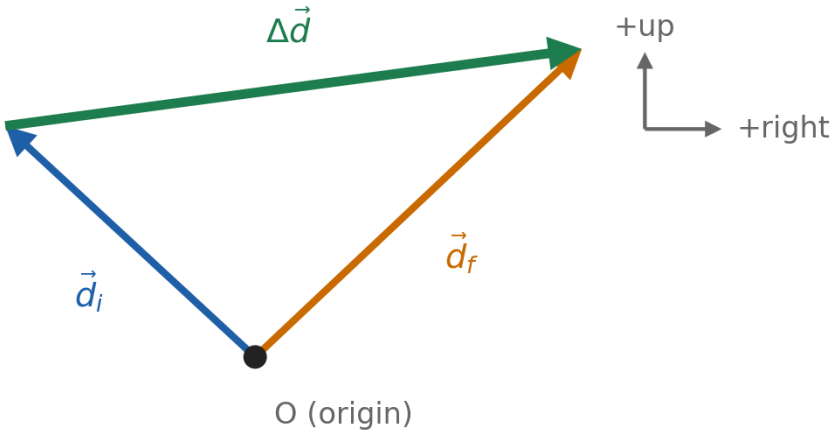
The position of a student at two points in time is shown by the vectors below. Draw a tip-to-tail diagram that would show $\vec{d}_f - \vec{d}_i$. What does this vector represent physically?

Picture the situation



A student is first at one spot near the school, then later at another spot. Both positions are measured from the school.

Positions from the origin O

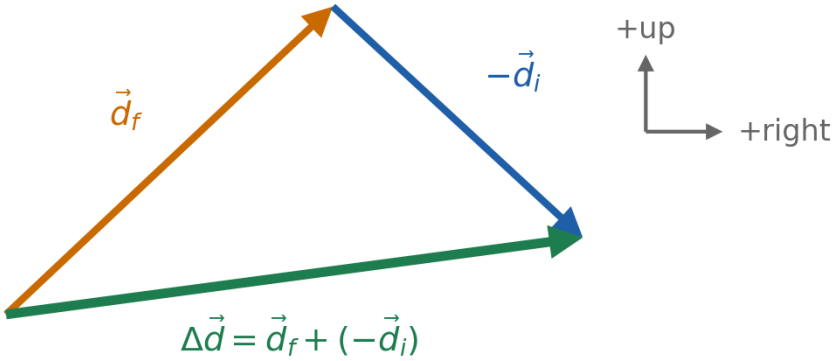


Step 1 – The tip-to-tail diagram

$\vec{d}_i$ is the student's first (initial) position and $\vec{d}_f$ is the later (final) position, both measured from O. As with velocities, rewrite the subtraction as adding the opposite. Draw $\vec{d}_f$, then from its tip draw $-\vec{d}_i$ ($\vec{d}_i$ reversed). The result goes from the first tail to the last tip.

$$\vec{d}_f - \vec{d}_i = \vec{d}_f + (-\vec{d}_i) = \Delta\vec{d}$$

Tip-to-tail: $\vec{d}_f + (-\vec{d}_i)$



Step 2 – What the arrow means

Compare with the position picture: the green arrow is the same as the arrow from the student's first position straight to the second position. That straight-line change in position is the displacement.

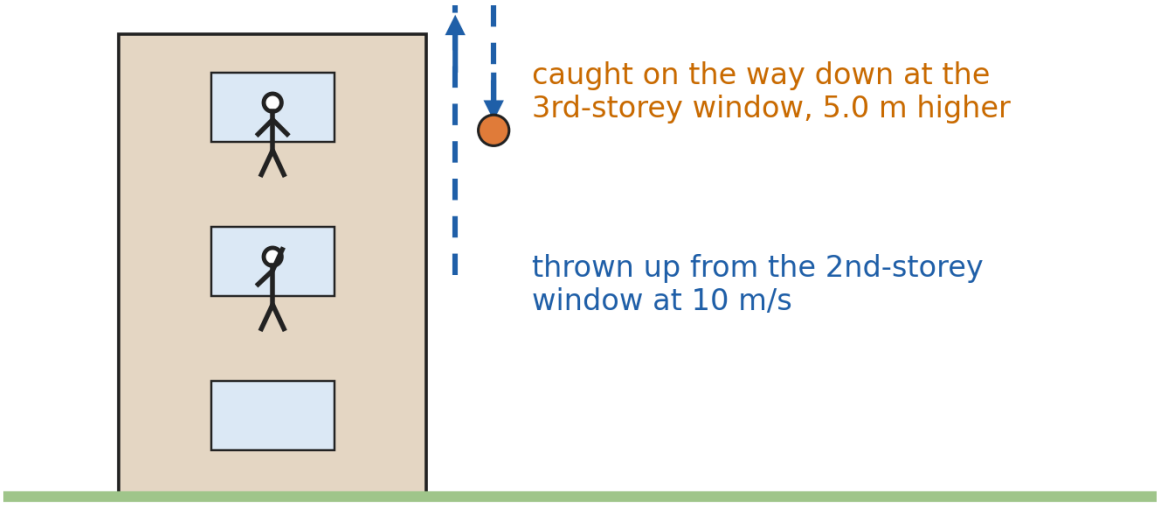
> **$\vec{d}_f - \vec{d}_i = \Delta\vec{d}$, the student's displacement (from the first position to the second)**

ADDITIONAL 1

Ball caught on the way down

A ball is thrown vertically upward from a 2nd story window with an initial velocity of 10 m/s. If a person in a third story window that is 5.0 m above the first window catches the ball on its way down, how long was the ball in the air?

Picture the situation



Someone throws a ball straight up out of a window. It rises a little past the window above, stops, and starts falling. A person at the upper window catches it as it comes back down past them.

SOLVING FOR

The time Δt from the throw until the catch.

GIVEN

$\vec{v}_1 = +10$ m/s

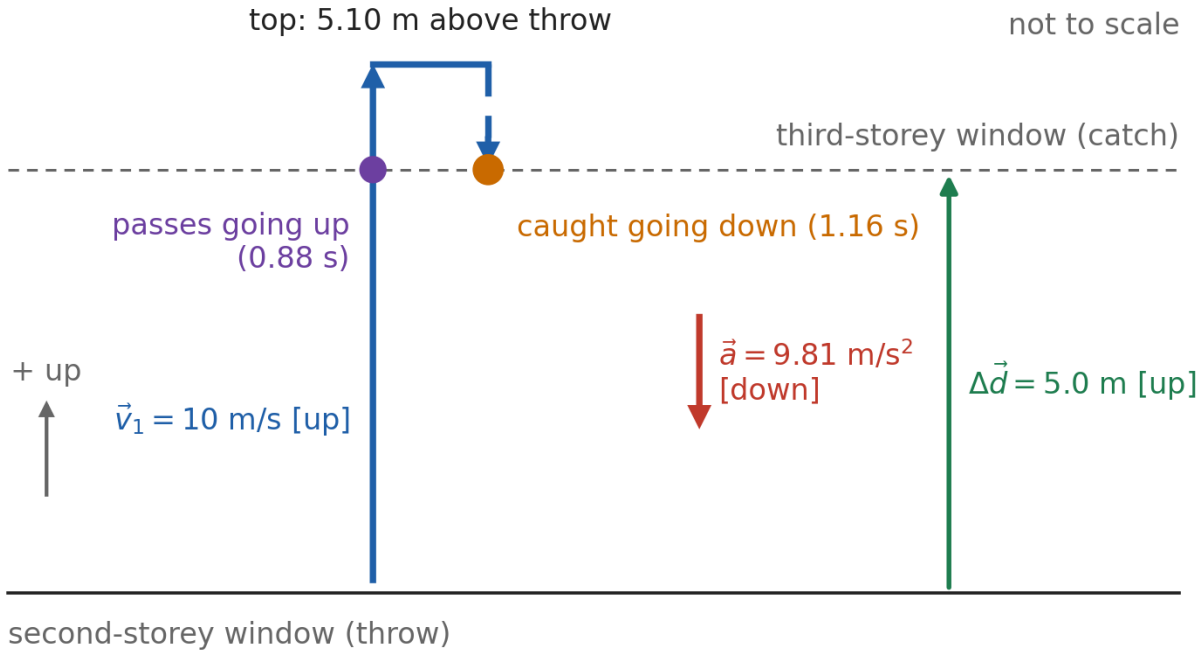
$\Delta\vec{d} = +5.0$ m

$\vec{a} = -9.81$ m/s^2

$\Delta t = ?$

Sign convention: up is positive. The catch is above the throw, so Δd is positive. Gravity points down, so a is negative.

Diagram: motion picture



Step 1 – Choose the equation

The ball moves only under gravity, so its acceleration is constant and the constant-acceleration equations apply. We know v_1, a and Δd, and we want Δt. We don't know (and don't need) the final velocity, so we pick the equation that doesn't contain v_2. Then substitute and move everything to one side.

Formula: $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2$

Put in the numbers: $5.0 = 10\,\Delta t + \frac{1}{2}(-9.81)\,\Delta t^2$

Simplify $\frac{1}{2}(-9.81) = -4.905$: $5.0 = 10\,\Delta t - 4.905\,\Delta t^2$

Move everything to the left side (add $4.905\Delta t^2$, subtract $10\Delta t$):

$4.905\,\Delta t^2 - 10\,\Delta t + 5.0 = 0$

Step 2 – Solve the quadratic

Δt appears both squared and on its own, so this is a quadratic equation and we can't just isolate Δt. We use the quadratic formula. Match the equation to $a\Delta t^2 + b\Delta t + c = 0$: a = 4.905 (the number in front of Δt^2), b = −10 (in front of Δt) and c = 5.0 (the number on its own).

Formula: $\Delta t = \dfrac{-b \pm \sqrt{b^2 - 4ac}}{2a}$

$$\Delta t = \frac{-(-10) \pm \sqrt{(-10)^2 - 4(4.905)(5.0)}}{2(4.905)}$$

$$\Delta t = \frac{10 \pm \sqrt{100 - 98.1}}{9.81} = \frac{10 \pm \sqrt{1.90}}{9.81} = \frac{10 \pm 1.378}{9.81}$$

With $-$: $\Delta t = \dfrac{8.622}{9.81} = 0.88$ s With $+$: $\Delta t = \dfrac{11.378}{9.81} = 1.16$ s

Step 3 – Pick the right answer

Why two answers? The ball is level with the upper window twice: once on the way up (0.88 s) and once on the way down (1.16 s). The question says it is caught on the way down, so we want the later time. As a check, find the velocity at 1.16 s:

Formula: $v_2 = v_1 + a\Delta t$

$$v_2 = 10 + (-9.81)(1.160) = 10 - 11.38 = -1.38 \text{ m/s}$$

The velocity is negative, so the ball is moving down at that moment. That confirms 1.16 s is the catch.

The ball was in the air for 1.2 s.

The ball only just gets past the upper window. At the top $v_2 = 0$, so $v_2^2 = v_1^2 + 2a\Delta d$ gives a highest point of $10^2/(2 \times 9.81) = 5.10$ m above the throw, only 0.10 m above the catch. That is why the two times are close together.

ADDITIONAL 2

Two opposite vectors

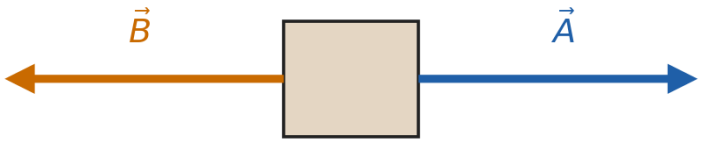
If the sum of two vectors equals zero, the magnitude of their difference is

a) equal to the magnitude of either one

b) equal to twice the magnitude of either one

c) less than twice the magnitude of either one.

Picture the situation



same size, opposite directions: they add to zero

Two arrows that add to zero must cancel exactly, like two equal pulls in opposite directions on a box.

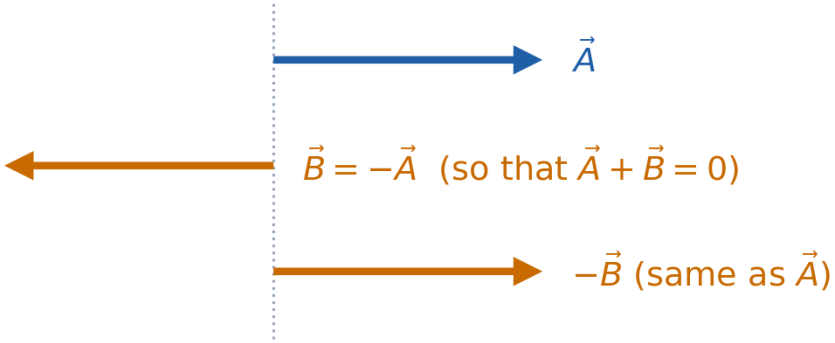
Step 1 – What “the sum is zero” tells us

If $\vec{A} + \vec{B} = 0$, then $\vec{B} = -\vec{A}$. So $\vec{B}$ is the same length as $\vec{A}$ but points the opposite way.

$$\vec{A} + \vec{B} = 0$$

Subtract $\vec{A}$ from both sides: $\vec{B} = -\vec{A}$

Diagram: the two vectors (tails lined up)



Step 2 – Work out the difference

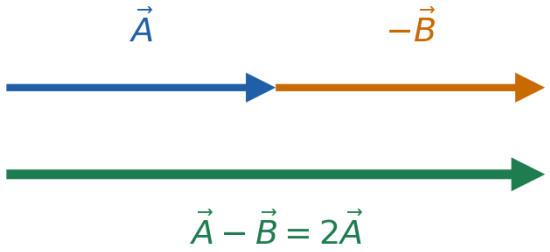
To subtract $\vec{B}$, add its opposite, $-\vec{B}$. But $-\vec{B}$ is the same as $\vec{A}$ (reversing $\vec{B}$ turns it back into $\vec{A}$). So the difference is $\vec{A}$ added to itself: two equal arrows pointing the same way, tip to tail, make one arrow twice as long.

$$\vec{A} - \vec{B} = \vec{A} + (-\vec{B})$$

Since $\vec{B} = -\vec{A}$, then $-\vec{B} = \vec{A}$: $\quad \vec{A} - \vec{B} = \vec{A} + \vec{A} = 2\vec{A}$

So its size is $\quad |\vec{A} - \vec{B}| = 2|\vec{A}|$

Diagram: $\vec{A}$ + $(-\vec{B})$, tip to tail



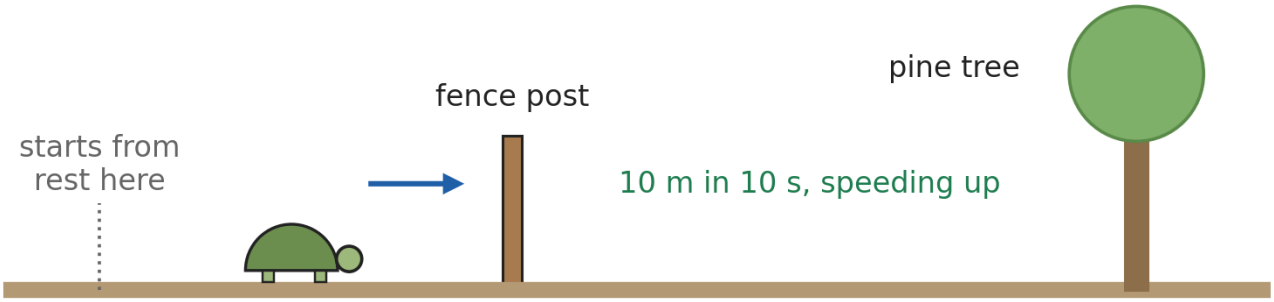
(b) equal to twice the magnitude of either one

ADDITIONAL 3

Turtle: find the rest starting point

A turtle is moving with constant acceleration along a straight ditch. He starts his stopwatch as he passes a fence post and notes that it takes him 10s to reach a pine tree 10m farther along the ditch. As he passes the pine tree his speed is 1.2 m/s. How far was he from the fence post when he started from rest? (SIN 76)

Picture the situation



The turtle started from rest somewhere before the fence post and has been speeding up steadily ever since. We know about the fence-to-tree part and must work backward to where he started.

SOLVING FOR

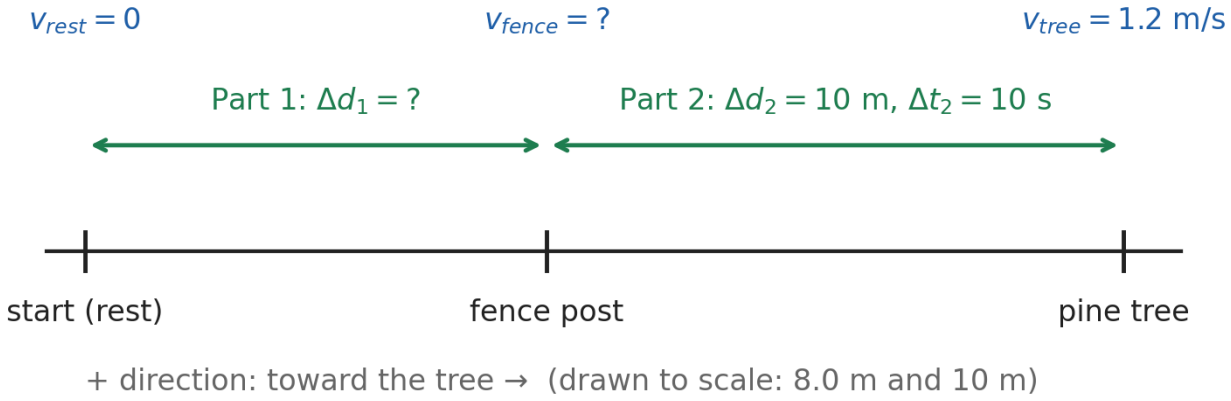
The distance from the starting point (at rest) to the fence post, Δd_1.

GIVEN

$\Delta d_2 = 10$ m	$\Delta t_2 = 10$ s	$v_{tree} = 1.2$ m/s
$v_{rest} = 0$	$a =$ constant	$\Delta d_1 = ?$

Sign convention: + is the direction the turtle moves (toward the tree). He never turns around, so speeds and distances can be used without signs. The trip has two parts. **Part 1** is from the start (at rest) to the fence; **Part 2** is from the fence to the tree. A subscript 1 or 2 on Δd or Δt says which part it belongs to. The speeds are named by place: $v_{rest} = 0$ at the start, v_{fence} at the fence post and v_{tree} at the tree.

Diagram: motion picture



Step 1 – Find his speed at the fence post

We split the trip into two parts and start with the part we know most about: fence to tree. There we know Δd_2, Δt_2 and the final speed. Because the acceleration is constant, the average velocity is halfway between the start and end speeds, so the equation $\Delta d = \frac{1}{2}(v_1 + v_2)\Delta t$ works. For Part 2, v_1 (the speed at the start of the part) is v_{fence} and v_2 (the speed at the end) is v_{tree}. The only unknown is v_{fence}.

Formula: $\Delta d = \dfrac{v_1 + v_2}{2}\,\Delta t$

Multiply both sides by 2: $2\Delta d = (v_1 + v_2)\,\Delta t$

Divide both sides by Δt: $\dfrac{2\Delta d}{\Delta t} = v_1 + v_2$

Subtract v_2 from both sides: $v_1 = \dfrac{2\Delta d}{\Delta t} - v_2$

For Part 2: $v_{fence} = \dfrac{2\Delta d_2}{\Delta t_2} - v_{tree}$

$$v_{fence} = \frac{2(10\text{ m})}{10\text{ s}} - 1.2\text{ m/s} = 2.0 - 1.2 = 0.80\text{ m/s}$$

Step 2 – Find the acceleration

Now we know both speeds on the fence-to-tree part, so the definition of acceleration gives a. This same acceleration applies before the fence too, because it is constant the whole time.

Formula: $a = \frac{v_2 - v_1}{\Delta t}$

For Part 2: $a = \frac{v_{tree} - v_{fence}}{\Delta t_2}$

$a = \frac{1.2 - 0.80}{10} = \frac{0.40}{10} = 0.040 \text{ m/s}^2$

Step 3 – Distance from the start to the fence

Now look at Part 1, from the start to the fence. Here v_1 is $v_{rest} = 0$, v_2 is $v_{fence} = 0.80$ m/s and a = 0.040 m/s². We don't know how long it took, so we use the equation without Δt and solve it for Δd, which here is Δd_1.

Formula: $v_2^2 = v_1^2 + 2a\Delta d$

Subtract v_1^2 from both sides: $v_2^2 - v_1^2 = 2a\Delta d$

Divide both sides by $2a$: $\Delta d = \frac{v_2^2 - v_1^2}{2a}$

For Part 1: $\Delta d_1 = \frac{v_{fence}^2 - v_{rest}^2}{2a}$

$\Delta d_1 = \frac{(0.80)^2 - 0^2}{2(0.040)} = \frac{0.64}{0.080} = 8.0 \text{ m}$

He started from rest 8.0 m before the fence post.

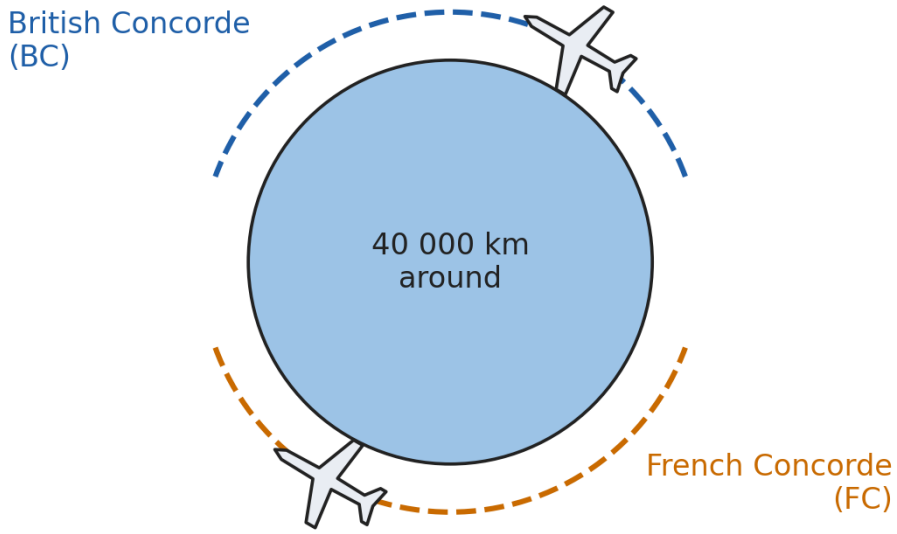
Check: from rest all the way to the tree: $(1.2)^2/(2 \times 0.040) = 18$ m, and 18 m − 10 m = 8.0 m.

ADDITIONAL 4

Concordes: equal distance versus equal time

A British Concorde (BC) and a French Concorde (FC) flew in opposite directions around the earth (40 000 km). The BC covered half of its flight distance at a supersonic speed of 2500 km/h and the other half at a subsonic speed of 1000 km/h. The FC spent half of its flight time at 2500 km/h and the other half at 1000 km/h. Which Concorde completed the trip first, and by how many hours did it beat the other?

Picture the situation



Two planes fly all the way around the earth in opposite directions. They use the same two speeds, but split their trips differently: BC by distance, FC by time.

GIVEN

$d = 40\,000$ km	$v_{fast} = 2500$ km/h	$v_{slow} = 1000$ km/h
$\Delta t_{BC} = ?$	$\Delta t_{FC} = ?$	

Sign convention: only speeds and distances are needed (each plane just goes around once), so no signs are used. d is a distance, v_{fast} and v_{slow} are the two speeds, and Δt_{BC} and Δt_{FC} are the total flight times of the British and French Concordes.

Step 1 – British Concorde (half the distance at each speed)

BC flies 20 000 km fast and 20 000 km slow. For each half, time = distance ÷ speed. Then add the two times.

Formula: $v = \frac{d}{\Delta t}$

Multiply both sides by Δt: $v\,\Delta t = d$

Divide both sides by v: $\Delta t = \frac{d}{v}$

$$\Delta t_{BC} = \frac{20\,000}{2500} + \frac{20\,000}{1000} = 8.0 + 20.0 = 28.0 \text{ h}$$

Step 2 – French Concorde (half the time at each speed)

We don't know FC's total time, Δt_{FC}, yet. It flies for $\Delta t_{FC}/2$ at each speed. The distance in each half is speed × time, and the two distances must add up to 40 000 km. That gives one equation with Δt_{FC} as the only unknown.

Formula: $d = v\,\Delta t$ (for each half)

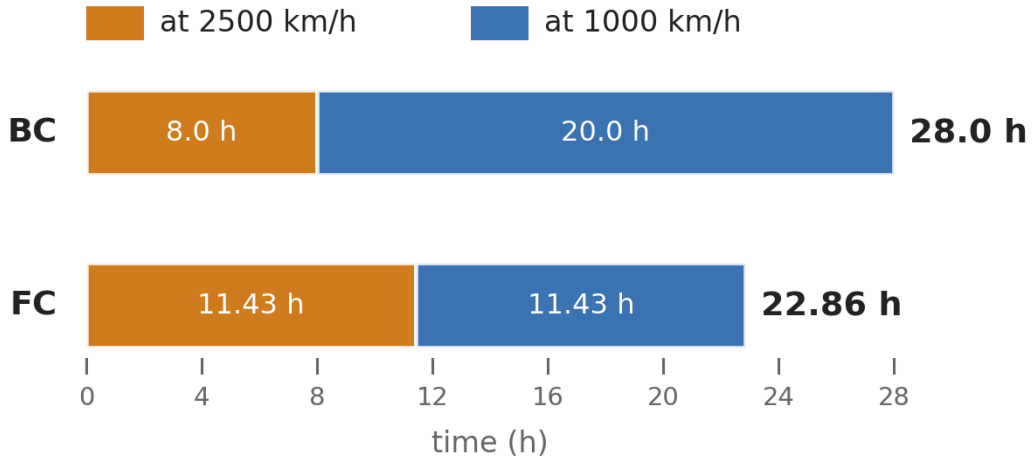
$$2500\left(\frac{\Delta t_{FC}}{2}\right) + 1000\left(\frac{\Delta t_{FC}}{2}\right) = 40\,000$$

Simplify each term: $1250\,\Delta t_{FC} + 500\,\Delta t_{FC} = 40\,000$

Add the like terms: $1750\,\Delta t_{FC} = 40\,000$

Divide both sides by 1750: $\Delta t_{FC} = \frac{40\,000}{1750} = 22.86 \text{ h}$

Diagram: how each plane spends its time (to scale)



Step 3 – Compare

The difference comes from how the time is spent. The slow half of BC's distance takes 20 of its 28 hours, so BC spends most of its time slow. FC spends exactly half its time fast. So FC wins.

$$\Delta t_{BC} - \Delta t_{FC} = 28.0 - 22.86 = 5.14 \text{ h}$$

The French Concorde finishes first, by 5.1 h.

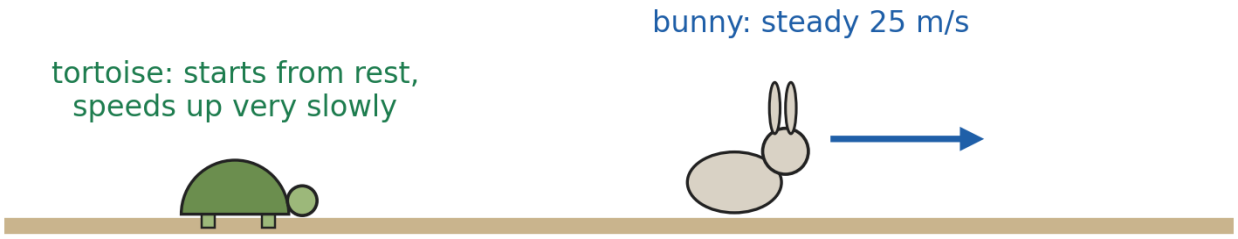
Check: each half lasts $\Delta t_{FC}/2 = 11.43$ h, so FC's distances are $2500 \times 11.43 \approx 28\,570$ km and $1000 \times 11.43 \approx 11\,430$ km, which add to 40 000 km.

ADDITIONAL 5

Bunny and accelerating tortoise

The Easter bunny runs along a straight and narrow path with a constant speed of 25m/s. He passes a sleeping tortoise, which immediately starts to chase the bunny with a constant acceleration of 3.0×10^{-3} m/s^2. How long does it take the tortoise to catch the bunny?

Picture the situation



The bunny zooms past at a steady speed. The tortoise wakes up and starts from rest, speeding up very slowly but steadily. Eventually it catches up.

GIVEN

$v_{bunny} = 25$ m/s

$v_{1(\text{tortoise})} = 0$

$a_{tortoise} = 3.0 \times 10^{-3}$ m/s^2

$\Delta t = ?$

Sign convention: + is the bunny's direction. Both start at the tortoise's position at $\Delta t = 0$. The word in each subscript says which animal a symbol belongs to, and Δt is the time since the bunny passed.

Step 1 – Write how far each one goes in time Δt

The bunny has constant velocity, so its distance is just speed × time. The tortoise starts from rest with constant acceleration, so we use $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2$ with the tortoise's values: $v_{1(\text{tortoise})} = 0$ and $a_{\text{tortoise}} = 3.0 \times 10^{-3}\ \text{m/s}^2$.

Formula (bunny, constant velocity): $\Delta d = v\,\Delta t \Rightarrow \Delta d_{bunny} = 25\,\Delta t$

Formula (tortoise): $\Delta d = v_1 \Delta t + \frac{1}{2} a \Delta t^2$

$$\Delta d_{tortoise} = (0)\Delta t + \frac{1}{2}(3.0 \times 10^{-3})\,\Delta t^2 = (1.5 \times 10^{-3})\,\Delta t^2$$

Step 2 – Set them equal

The tortoise catches the bunny when both have gone the same distance from the same start. So we set the two expressions equal. Both sides have a Δt, and we can divide it out because we want the later meeting, not Δt = 0 (the moment they start together).

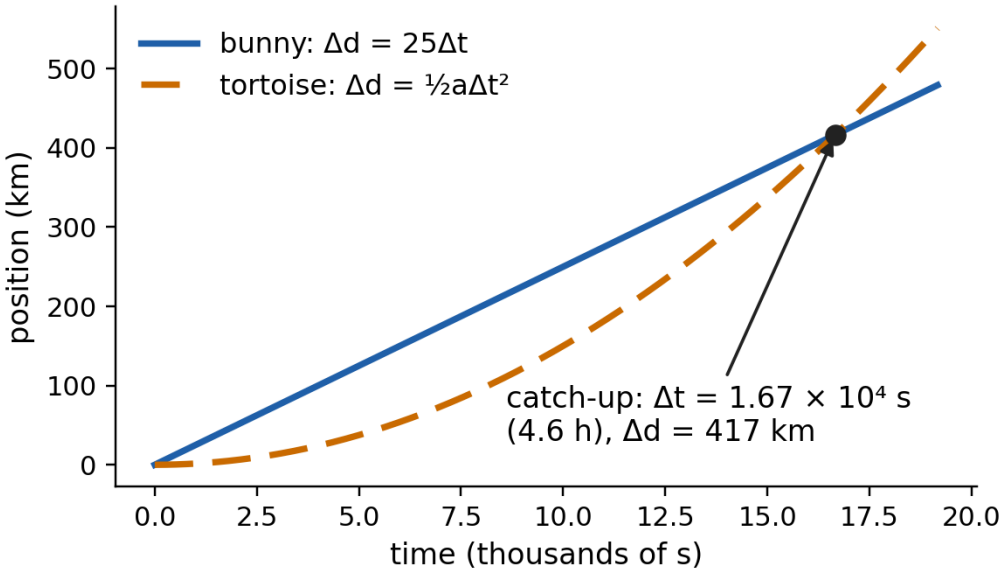
$$25\,\Delta t = (1.5 \times 10^{-3})\,\Delta t^2$$

Divide both sides by Δt: $25 = (1.5 \times 10^{-3})\,\Delta t$

Divide both sides by 1.5×10^{-3}: $\Delta t = \dfrac{25}{1.5 \times 10^{-3}} = 16\,667\ \text{s}$

$16\,667\ \text{s} \div 3600\ \text{s/h} = 4.63\ \text{h}$

Graph: position vs time for both animals



The tortoise catches the bunny after 1.7×10^4 s (about 4.6 h).

Check: at the catch the tortoise is moving at aΔt = 50 m/s, twice the bunny's speed. That makes sense: it started at 0, so to cover the same distance in the same time its average speed must equal the bunny's 25 m/s.

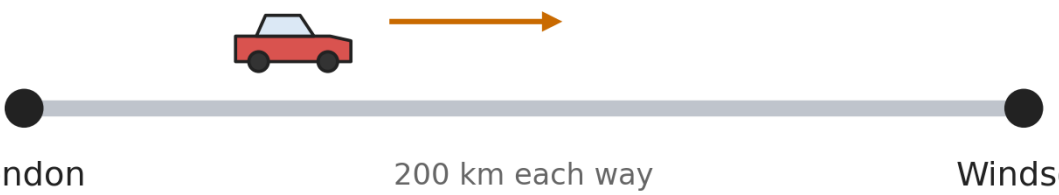
ADDITIONAL 6

Required round-trip average speed

A person travelling between London and Windsor (200 km apart) wants to complete a round trip at an average speed of 100 km/h. On arriving at Windsor at the end of the first leg, she discovers that she has only averaged 75km/h. What must her average speed by on the return trip? (CAP Contest 1984)

Picture the situation

there at 75 km/h, back at ? — average for the round trip must be 100 km/h



She drives 200 km to Windsor and 200 km back. The slow first leg has used up some of her time, so the return leg must be faster.

GIVEN

$d = 200$ km each way $\qquad v_{ave,\,trip} = 100$ km/h

$v_{ave,\,out} = 75$ km/h $\qquad v_{ave,\,back} = ?$

Sign convention: average speed = total distance ÷ total time (speeds, so no directions). “Out” is the first leg (London to Windsor), “back” is the return leg, and “trip” means the whole round trip. So $v_{ave,\,out}$ is her average speed on the first leg and Δt_{back} is the time she has left for the return.

Step 1 – How much time is allowed for the whole trip

The trick is to think in terms of time, not speed. To average 100 km/h over the 400 km round trip, the whole trip must take:

Formula: $v_{ave} = \dfrac{d}{\Delta t}$

Multiply both sides by Δt: $\quad v_{ave}\,\Delta t = d$

Divide both sides by v_{ave}: $\quad \Delta t = \dfrac{d}{v_{ave}}$

$$\Delta t_{trip} = \frac{400\text{ km}}{100\text{ km/h}} = 4.00\text{ h}$$

Step 2 – How much time is left for the return

The first leg used 200 km ÷ 75 km/h of that time. Subtract to find what is left.

$$\Delta t_{out} = \frac{200\text{ km}}{75\text{ km/h}} = 2.67\text{ h}$$

$$\Delta t_{back} = 4.00 - 2.667 = 1.33\text{ h}$$

Diagram: the 4.00 h time budget



Whole trip must take 400 km ÷ 100 km/h = 4.00 h

Step 3 – Speed needed on the return

She must cover 200 km in the 1.33 h that is left.

Formula: $v_{ave} = \dfrac{d}{\Delta t}$

$$v_{ave,\,back} = \frac{200\ \text{km}}{1.333\ \text{h}} = 150\ \text{km/h}$$

She must average 150 km/h on the return trip.

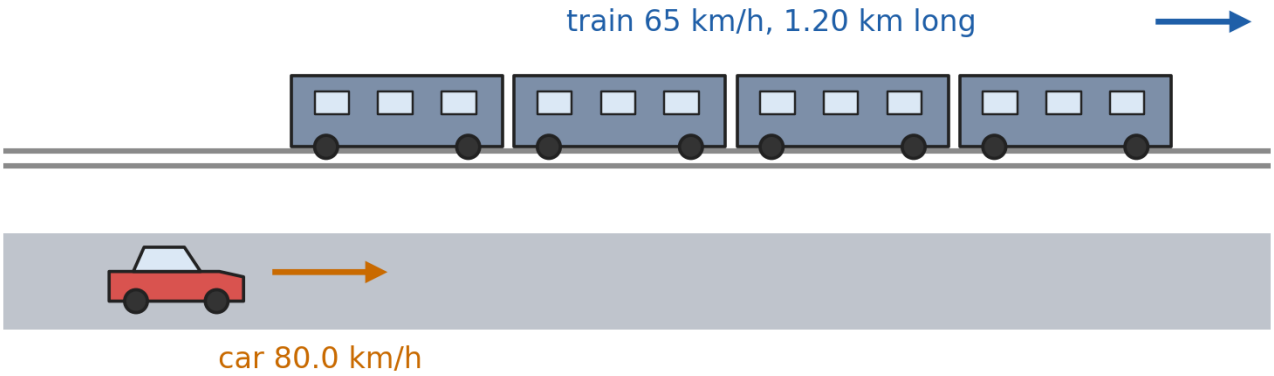
A common wrong answer is 125 km/h (so the two speeds “average” to 100). That fails because she spends more time on the slow leg than on the fast leg, and average speed depends on time, not on the number of legs.

ADDITIONAL 7

Car passing a train

A car going 80.0 km/h overtakes a 1.20 km long train that is travelling at 65 km/h in the same direction on a track parallel to the road. How long does it take the car to pass the train? How long would it take if the vehicles were approaching each other travelling in opposite directions?

Picture the situation



A car on a road beside a railway catches up to a long train and drives past it.

GIVEN

$\vec{v}_{CG} = +80.0$ km/h $|\vec{v}_{TG}| = 65$ km/h $L = 1.20$ km

$\Delta t = ?$ (each case)

Sign convention: + is the car's direction. Subscripts: C = car, T = train, G = ground; v_{CT} is the car's velocity relative to the train. The car's own length is ignored.

Diagram: what "passing" means

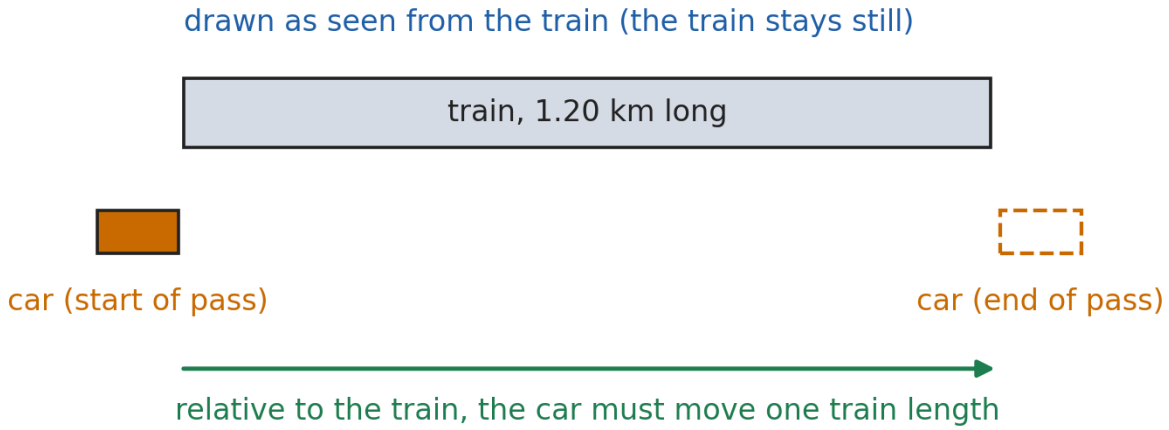
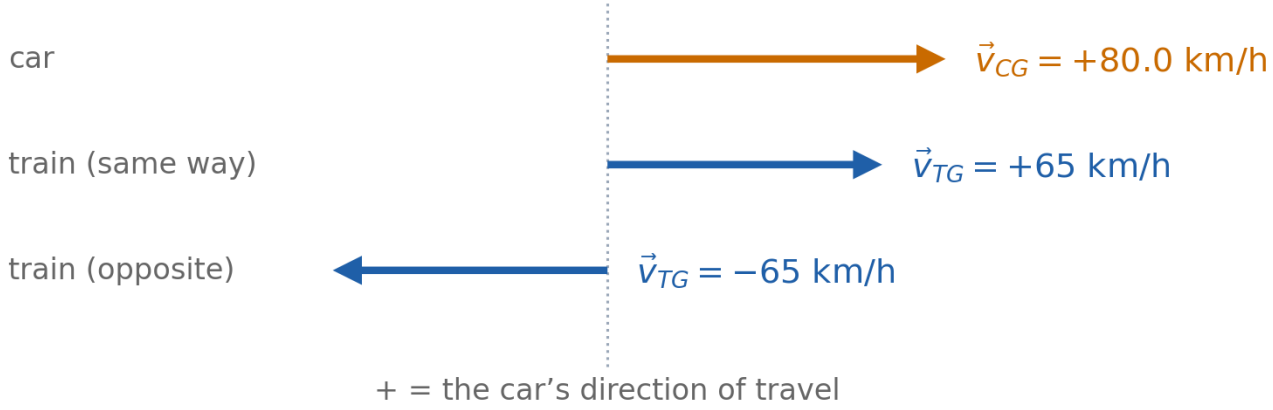


Diagram: velocities relative to the ground



Step 1 – Look at it from the train

The easiest way is to imagine sitting on the train. From there, the train isn't moving, and the car only has to travel one train length, L = 1.20 km, to get from one end of the train to the other. Its velocity as seen from the train, $\vec{v}_{CT}$, is its ground velocity minus the train's ground velocity.

$$\vec{v}_{CT} = \vec{v}_{CG} - \vec{v}_{TG}$$

Step 2 – Same direction

Both move forward, so the car only gains 15 km/h on the train. Time = distance ÷ speed, where the distance is the train length L and the speed is v_{CT}.

$$\vec{v}_{CT} = 80.0 - 65 = +15 \text{ km/h}$$

Formula: $v = \frac{d}{\Delta t}$

Multiply both sides by Δt: $v\,\Delta t = d$

Divide both sides by v: $\Delta t = \frac{d}{v}$

Here $d = L$ and $v = v_{CT}$: $\Delta t = \frac{L}{v_{CT}}$

$$\Delta t = \frac{1.20 \text{ km}}{15 \text{ km/h}} = 0.0800 \text{ h}$$

Convert to minutes (1 h = 60 min): $0.0800 \times 60 = 4.80$ min

Step 3 – Opposite directions

Now the train moves toward the car, so its velocity is −65 km/h. Subtracting a negative adds the speeds: they close the gap much faster.

$$\vec{v}_{CT} = 80.0 - (-65) = 80.0 + 65 = +145 \text{ km/h}$$

Formula: $\Delta t = \frac{L}{v_{CT}}$

$$\Delta t = \frac{1.20 \text{ km}}{145 \text{ km/h}} = 0.008276 \text{ h}$$

Convert (1 h = 3600 s):

$$0.008276 \times 3600 = 29.8 \text{ s} \quad (= 0.497 \text{ min})$$

Same direction: 4.8 min

Opposite directions: 29.8 s (0.497 min)

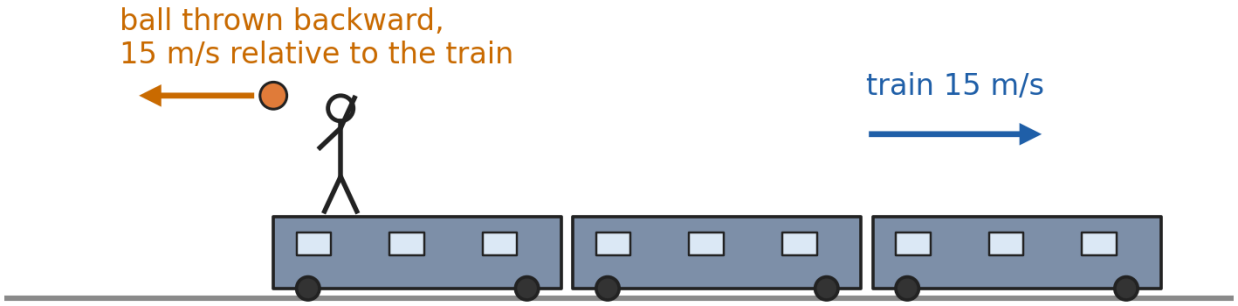
Significant figures: 80.0 − 65 = 15 has two, so the first answer has two (4.8 min). 80.0 + 65 = 145 has three, so the second has three.

RELATIVE 1

Baseball thrown backward on a train

A passenger at the rear of a train traveling at 15 m/s relative to Earth throws a baseball with a speed of 15 m/s in the direction opposite the motion of the train. What is the velocity of the baseball relative to Earth as it leaves the thrower's hand?

Picture the situation



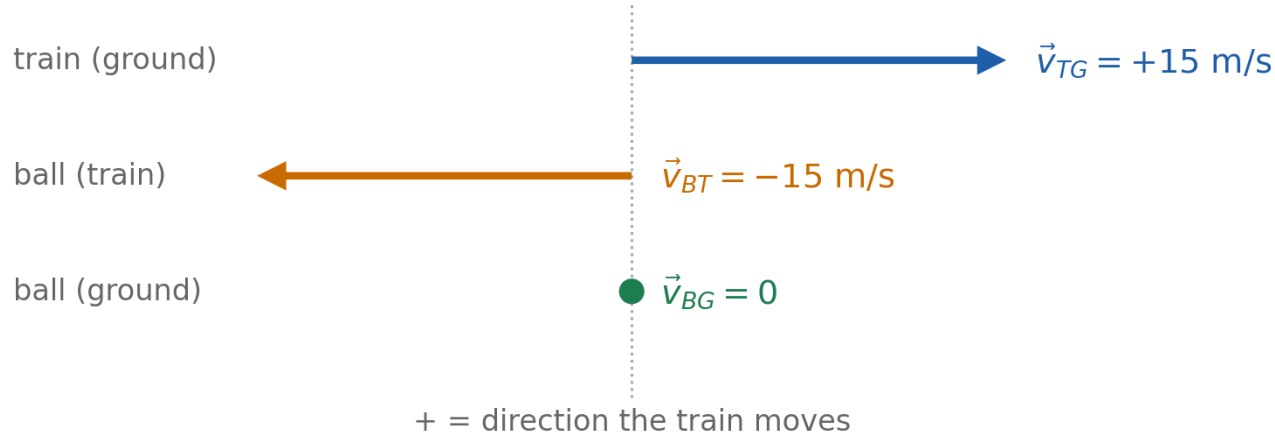
A passenger at the back of a moving train throws a ball backward, out over the tracks, at the same speed the train is moving forward.

GIVEN

$\vec{v}_{TG} = +15$ m/s $\qquad \vec{v}_{BT} = -15$ m/s $\qquad \vec{v}_{BG} = ?$

Sign convention: + is the train's direction. B = ball, T = train, G = ground (Earth). Relative velocity: $\vec{v}_{XZ} = \vec{v}_{XY} + \vec{v}_{YZ}$, where X, Y and Z stand for any three objects (they are not the letters of this question). The inner letters match and "cancel": X relative to Y, plus Y relative to Z, gives X relative to Z. Also $\vec{v}_{YX} = -\vec{v}_{XY}$ (switching the order of the letters reverses the direction).

Diagram: the velocities, one per row (tails lined up)



Step 1 – Add the relative velocities

We want the ball relative to the ground, but we know the ball relative to the train and the train relative to the ground. Chaining them (ball → train → ground) gives what we want. The train carries the ball forward at 15 m/s while the throw sends it backward at 15 m/s, so the two cancel.

Formula: $\vec{v}_{XZ} = \vec{v}_{XY} + \vec{v}_{YZ}$

$$\vec{v}_{BG} = \vec{v}_{BT} + \vec{v}_{TG} = (-15) + (+15) = 0 \text{ m/s}$$

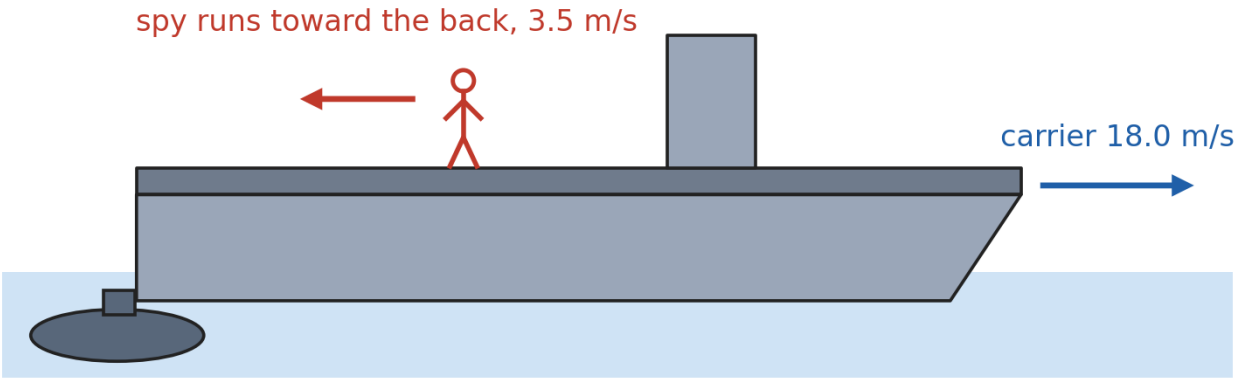
0 m/s: as it leaves the hand the ball is momentarily at rest relative to Earth (then it simply falls).

RELATIVE 2

Spy on an aircraft carrier

A spy runs from the front to the back of an aircraft carrier at a velocity of 3.5 m/s. If the aircraft carrier is moving forward at 18.0 m/s, how fast does the spy appear to be running when viewed by an observer on a nearby stationary submarine?

Picture the situation



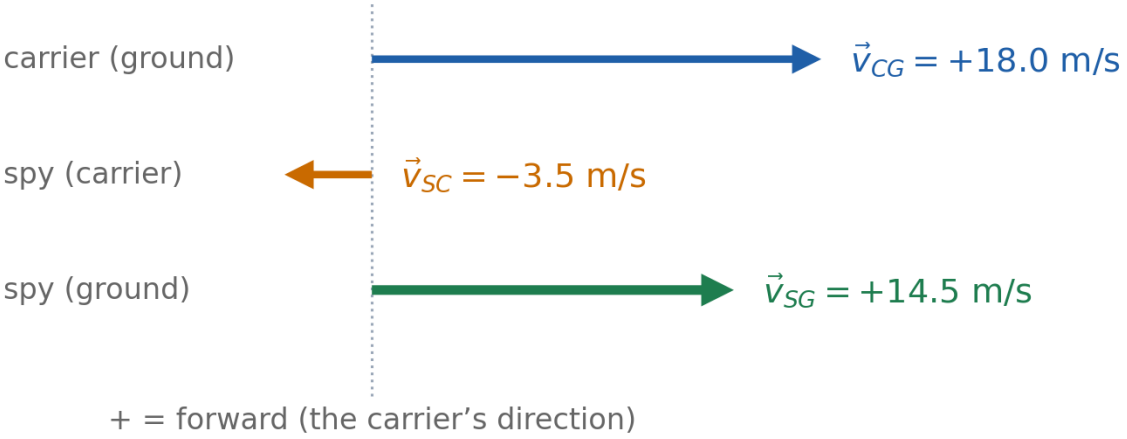
The carrier steams forward while the spy runs toward its back. The submarine is at rest in the water, so it sees velocities relative to the ground (Earth).

GIVEN

$\vec{v}_{CG} = +18.0$ m/s $\quad \vec{v}_{SC} = -3.5$ m/s $\quad \vec{v}_{SG} = ?$

Sign convention: + is the carrier's forward direction. S = spy, C = carrier, G = ground (the submarine is at rest, so it sees ground velocities). Running toward the back is negative.

Diagram: the velocities, one per row (tails lined up)



Step 1 – Add the relative velocities

Chain spy → carrier → ground. The spy runs backward on the deck, but the deck itself is moving forward much faster, so overall he still moves forward, just a bit slower than the ship.

Formula: $\vec{v}_{XZ} = \vec{v}_{XY} + \vec{v}_{YZ}$

$\vec{v}_{SG} = \vec{v}_{SC} + \vec{v}_{CG} = (-3.5) + (+18.0) = +14.5$ m/s

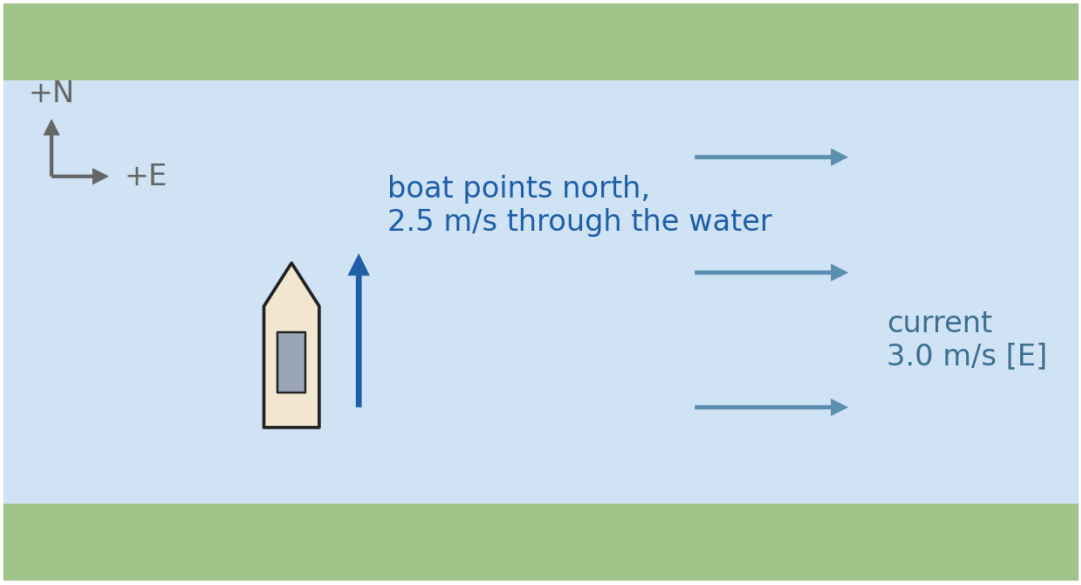
14.5 m/s, moving forward (in the carrier's direction), even though he is running toward the back.

RELATIVE 3

Ferry crossing a river

A ferry is crossing a river. If the ferry is headed due north with a speed of 2.5 m/s relative to the water and the river's velocity is 3.0 m/s to the east, what will the boat's velocity relative to Earth be? (Hint: Remember to include the direction in describing the velocity.)

Picture the situation



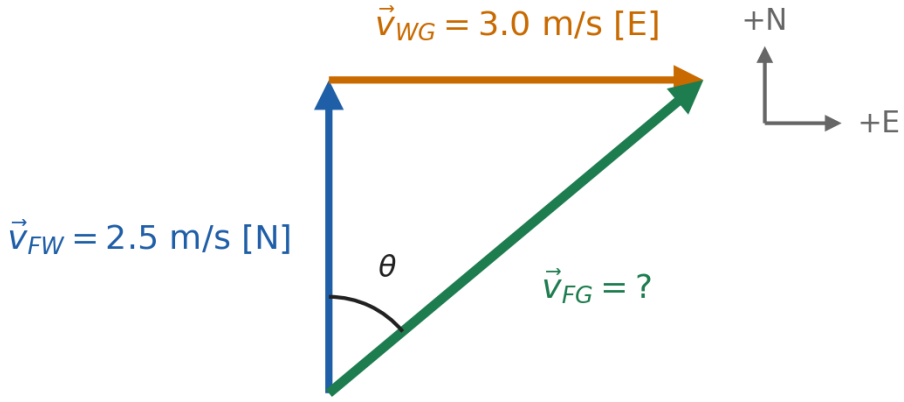
Seen from above: the ferry points straight across (north), but the current carries it east at the same time, so it crosses on a slant.

GIVEN

$\vec{v}_{FW} = 2.5$ m/s [N] $\quad \vec{v}_{WG} = 3.0$ m/s [E] $\quad \vec{v}_{FG} = ?$

Sign convention: F = ferry, W = water, G = ground (Earth). Relative velocity: $\vec{v}_{XZ} = \vec{v}_{XY} + \vec{v}_{YZ}$ (see Relative 1).

Diagram: $\vec{v}_{FW}$ + $\vec{v}_{WG}$, tip to tail



Step 1 – Add the two velocities

The boat's velocity relative to the ground is its velocity through the water plus the water's velocity relative to the ground (boat → water → ground). Draw them tip to tail. North and east are perpendicular, so the triangle has a right angle: Pythagoras gives the size, and tangent gives θ measured from north toward east (opposite side = 3.0 m/s east, adjacent side = 2.5 m/s north).

Formula (Pythagoras): $c = \sqrt{a^2 + b^2}$

$$|\vec{v}_{FG}| = \sqrt{2.5^2 + 3.0^2} = \sqrt{6.25 + 9.00} = \sqrt{15.25} = 3.91 \text{ m/s}$$

Formula: $\theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right)$

$$\theta = \tan^{-1}\left(\frac{3.0}{2.5}\right) = 50.2° \text{ east of north}$$

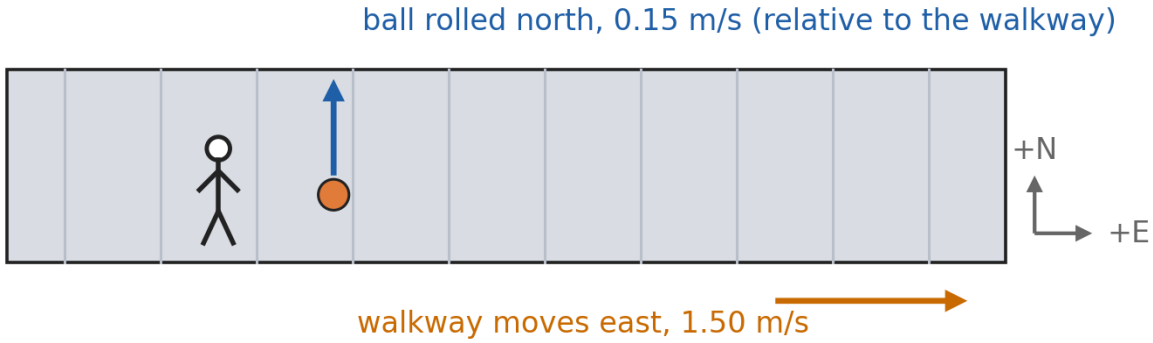
$\vec{v}_{FG}$ = 3.9 m/s [N 50° E]

RELATIVE 4

Ball on a moving walkway

A girl at an airport rolls a ball north on a moving walkway that moves east. If the ball's speed with respect to the walkway is 0.15 m/s and the walkway moves at a speed of 1.50 m/s, what is the velocity of the ball relative to the ground?

Picture the situation



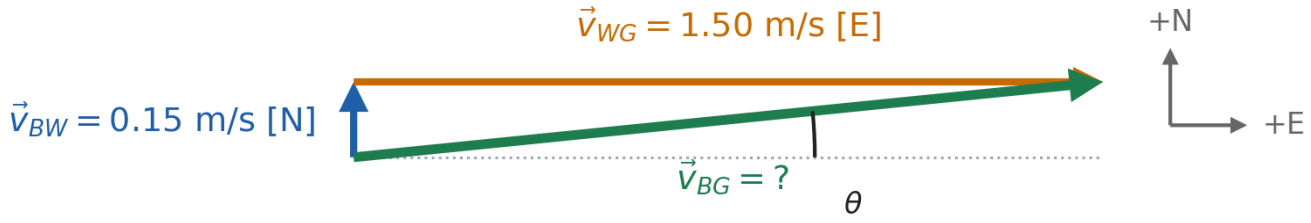
Seen from above: the girl rolls the ball across the walkway (north) while the walkway carries everything on it east.

GIVEN

$\vec{v}_{BW} = 0.15$ m/s [N] $\quad \vec{v}_{WG} = 1.50$ m/s [E] $\quad \vec{v}_{BG} = ?$

Sign convention: B = ball, W = walkway, G = ground. Relative velocity: $\vec{v}_{XZ} = \vec{v}_{XY} + \vec{v}_{YZ}$ (see Relative 1).

Diagram: $\vec{v}_{BW} + \vec{v}_{WG}$, tip to tail (to scale)



Step 1 – Add the two velocities

Ball → walkway → ground: add the ball's velocity on the walkway to the walkway's velocity. They are perpendicular, so it is a right triangle again. This time measure θ from east toward north, because the walkway's 1.50 m/s is much bigger and the result points almost east.

Formula (Pythagoras): $c = \sqrt{a^2 + b^2}$

$$|\vec{v}_{BG}| = \sqrt{1.50^2 + 0.15^2} = \sqrt{2.25 + 0.0225} = \sqrt{2.2725} = 1.51 \text{ m/s}$$

Formula: $\theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right)$

$$\theta = \tan^{-1}\left(\frac{0.15}{1.50}\right) = 5.71° \text{ north of east}$$

$\vec{v}_{BG}$ = 1.5 m/s [E 5.7° N]

RELATIVE 5

Four frames of reference

Describe the motion of the following objects if they are observed from the stated frames of reference:

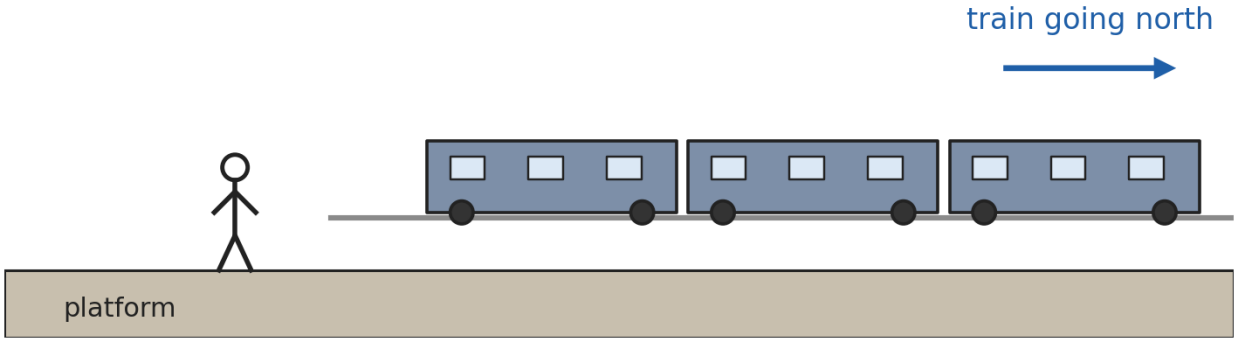
a. a person standing on a platform viewed from a train traveling north

b. a train traveling north viewed by a person standing on a platform

c. a ball dropped by a boy walking at a speed of 1 m/s viewed by the boy

d. a ball dropped by a boy walking 1 m/s as seen by a nearby viewer who is stationary

Picture the situation



A person stands on a platform as a train goes by heading north. Parts (c) and (d) are about a ball dropped by a walking boy.

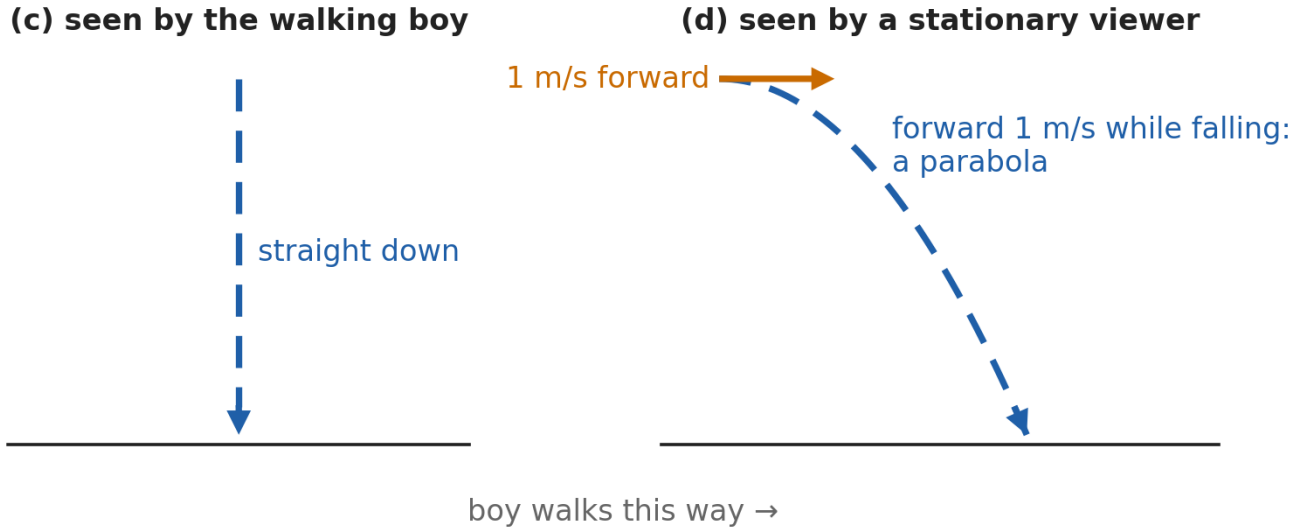
(a) and (b) Train and platform

Imagine sitting on the train looking out the window. You feel still, and the platform and everything on it seems to slide backward past you. In general, if X moves at some velocity relative to Y, then Y moves at the same speed in the opposite direction relative to X. Here that means $\vec{v}_{PT} = -\vec{v}_{TP}$.

$$\vec{v}_{PT} = -\vec{v}_{TP} \quad (P = \text{platform},\ T = \text{train})$$

(a) The person moves **south** at the train's speed. (b) The train moves **north** at its speed.

Diagram: the dropped ball in two frames



(c) and (d) The dropped ball

Before he lets go, the ball is moving forward with the boy at 1 m/s. Dropping it doesn't stop that forward motion. With no air resistance, nothing pushes the ball sideways, so it keeps moving forward at 1 m/s, exactly as fast as the boy. Meanwhile gravity pulls it down. So the ball stays right beside the boy (horizontally) the whole way down.

(c) The boy sees it fall **straight down**, speeding up. (d) The stationary viewer sees it move forward at 1 m/s while falling: a **parabola** (projectile motion).

(a) moves south at the train's speed

(b) moves north at the train's speed

(c) falls straight down

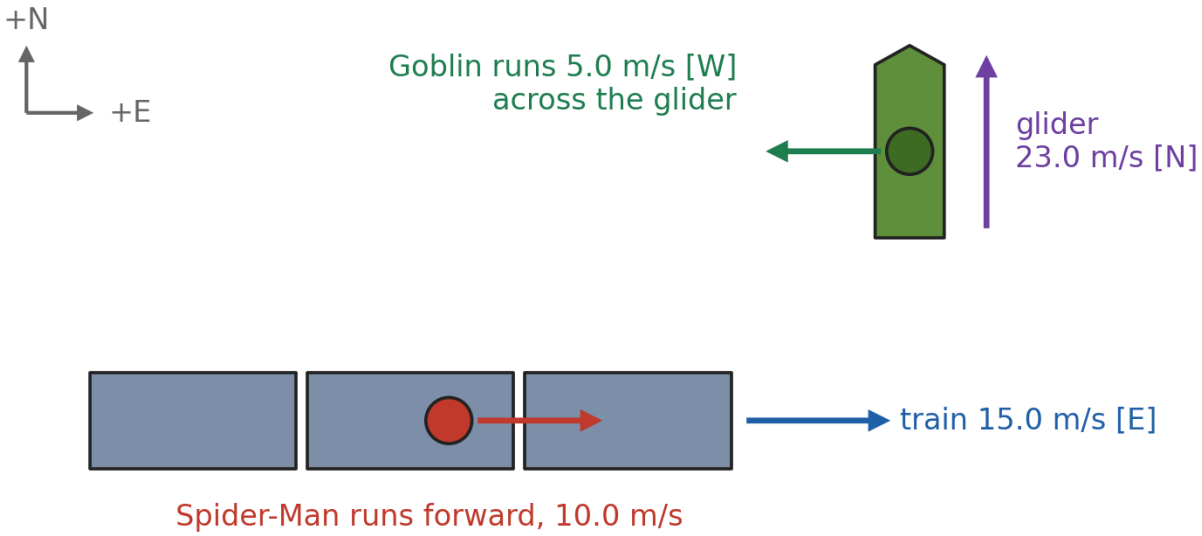
(d) follows a parabola, moving forward at 1 m/s as it falls

RELATIVE 6

Goblin relative to Spider-Man

If the Green Goblin is riding his Goblin Glider in an effort to evade Spiderman. Spiderman is running at 10.0 m/s toward the front on top of a train traveling at 15.0 m/s East. Green Goblin decides to run at 5.0 m/s W across his glider that is moving at 23.0 m/s North in an attempt to jump off and avoid Spiderman's web. What is Green Goblin's speed relative to Spiderman so Spiderman can accurately shot his web?

Picture the situation



Seen from above: Spider-Man runs forward along the top of an eastbound train. The Goblin runs west across a glider that is flying north.

GIVEN

$\vec{v}_{ST} = 10.0$ m/s [E]

$\vec{v}_{TE} = 15.0$ m/s [E]

$\vec{v}_{GL} = 5.0$ m/s [W]

$\vec{v}_{LE} = 23.0$ m/s [N]

$\vec{v}_{GS} = ?$

Sign convention: S = Spider-Man, T = train, G = Green Goblin, L = gLider, E = Earth (the ground). "Toward the front" of an eastbound train is east. Relative velocity: $\vec{v}_{XZ} = \vec{v}_{XY} + \vec{v}_{YZ}$ (see Relative 1).

Step 1 – Spider-Man relative to the ground

We want the Goblin relative to Spider-Man, but both are on moving things. The plan is to find each one's velocity relative to the ground first, then compare them. Spider-Man runs east on a train moving east, so the two velocities point the same way and simply add.

Formula: $\vec{v}_{XZ} = \vec{v}_{XY} + \vec{v}_{YZ}$

$\vec{v}_{SE} = \vec{v}_{ST} + \vec{v}_{TE} = 10.0 + 15.0 = 25.0$ m/s [E]

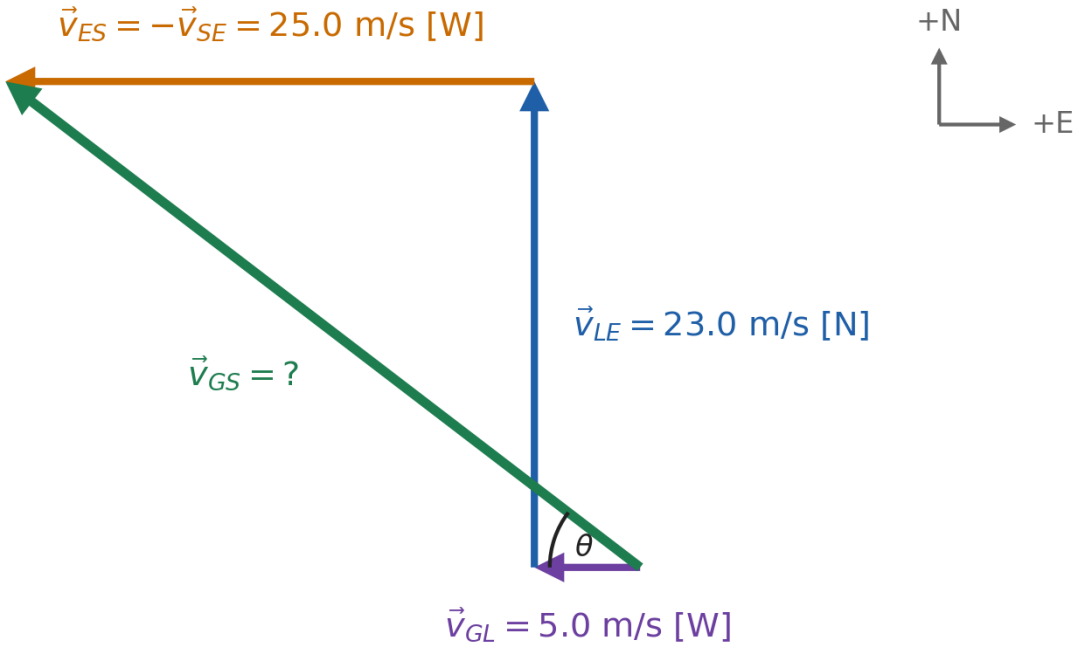
Step 2 – Goblin relative to Spider-Man

To see the Goblin from Spider-Man's point of view, chain the frames: Goblin → glider → Earth → Spider-Man. The last link is the Earth as seen by Spider-Man, which is the reverse of Spider-Man's velocity relative to the Earth: $\vec{v}_{ES} = -\vec{v}_{SE} = 25.0$ m/s [W]. Add the west parts together and the north parts together.

$\vec{v}_{GS} = \vec{v}_{GL} + \vec{v}_{LE} + \vec{v}_{ES} = \vec{v}_{GL} + \vec{v}_{LE} + (-\vec{v}_{SE})$

west: $5.0 + 25.0 = 30.0$ m/s north: 23.0 m/s

Diagram: the three vectors tip to tail



Step 3 – Size and direction

The west total and the north total are perpendicular, so they form a right triangle. Pythagoras gives the speed. For the direction, measure θ from west toward north: opposite side 23.0 (north), adjacent side 30.0 (west).

Formula (Pythagoras): $c = \sqrt{a^2 + b^2}$

$|\vec{v}_{GS}| = \sqrt{30.0^2 + 23.0^2} = \sqrt{900 + 529} = \sqrt{1429} = 37.8 \text{ m/s}$

Formula: $\theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right)$

$\theta = \tan^{-1}\left(\frac{23.0}{30.0}\right) = 37.48°$ north of west

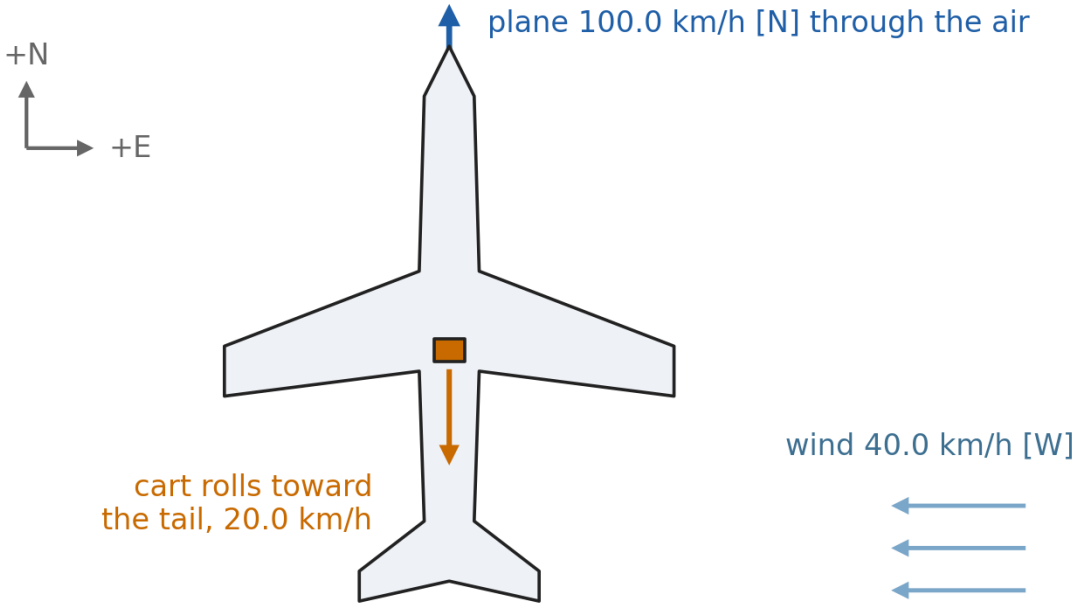
Speed 38 m/s (velocity 38 m/s [W 37° N] relative to Spider-Man)

RELATIVE 7

Beverage cart on a plane

A plane traveling north at 100.0 km/h through the air gets caught in a 40.0 km/h crosswind blowing west. This turbulence caused a beverage cart to brake free and begin rolling at 20.0 km/h toward the tail of the plane. What is the velocity of the cart relative to the ground? (you do not have to convert these since they are all the same unit)

Picture the situation



Seen from above: the plane points north through the air, the wind carries the air (and the plane) west, and inside the plane the cart rolls toward the tail.

GIVEN

$\vec{v}_{PA} = 100.0$ km/h [N]

$\vec{v}_{AG} = 40.0$ km/h [W]

$\vec{v}_{CP} = 20.0$ km/h [S]

$\vec{v}_{CG} = ?$

Sign convention: C = cart, P = plane, A = air, G = ground. The plane points north, so "toward the tail" is south.

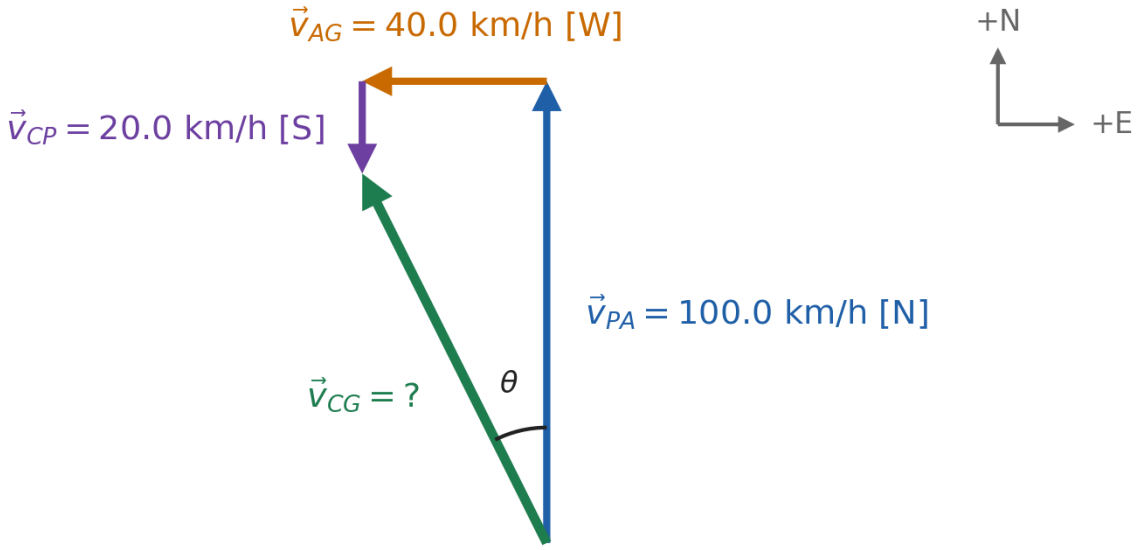
Step 1 – Chain the frames

Three things move here: the cart moves relative to the plane, the plane moves relative to the air, and the air (wind) moves relative to the ground. Chain them: cart → plane → air → ground. The cart rolls toward the tail (south) at 20.0 km/h while the plane carries it north at 100.0 km/h, so its north part is 80.0 km/h. The wind adds 40.0 km/h west.

$$\vec{v}_{CG} = \vec{v}_{CP} + \vec{v}_{PA} + \vec{v}_{AG}$$

north: $100.0 - 20.0 = 80.0$ km/h west: 40.0 km/h

Diagram: the three vectors tip to tail



Step 2 – Size and direction

North and west are perpendicular, so use Pythagoras for the speed and tangent for the angle, measured from north toward west (opposite 40.0, adjacent 80.0).

Formula (Pythagoras): $c = \sqrt{a^2 + b^2}$

$$|\vec{v}_{CG}| = \sqrt{80.0^2 + 40.0^2} = \sqrt{6400 + 1600} = \sqrt{8000} = 89.4 \text{ km/h}$$

Formula: $\theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right)$

$$\theta = \tan^{-1}\left(\frac{40.0}{80.0}\right) = 26.6° \text{ west of north}$$

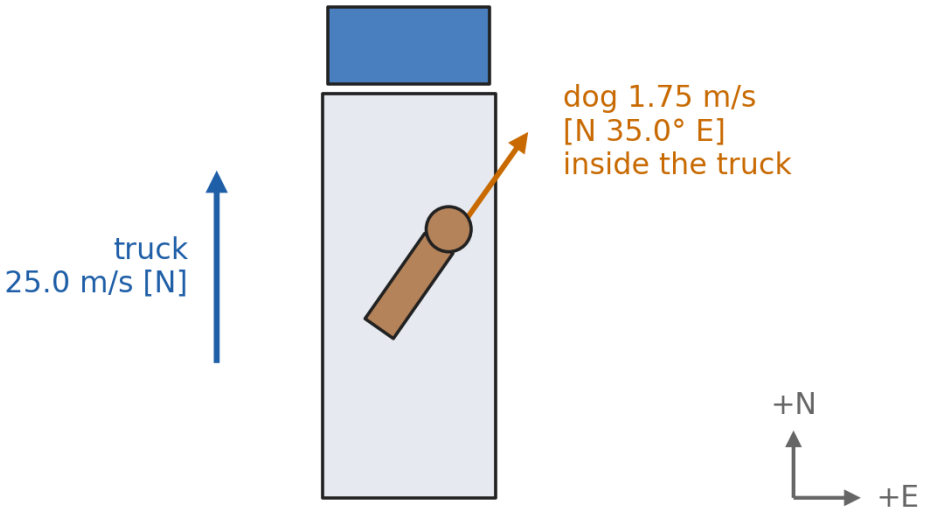
$\vec{v}_{CG}$ = 89.4 km/h [N 26.6° W]

RELATIVE 8 (HONOURS)

Dog walking inside a truck

A pet-store supply truck moves at 25.0 m/s north along a highway. Inside, a dog moves at 1.75 m/s at an angle of N 35.0° E. What is the velocity of the dog relative to the road?

Picture the situation



Seen from above: the truck drives north while the dog walks inside it, angled a little toward the east.

GIVEN

$\vec{v}_{TG} = 25.0$ m/s [N]

$\vec{v}_{DT} = 1.75$ m/s [N 35.0° E]

$\vec{v}_{DG} = ?$

Sign convention: D = dog, T = truck, G = ground (road). Relative velocity: $\vec{v}_{XZ} = \vec{v}_{XY} + \vec{v}_{YZ}$ (see Relative 1).

Step 1 – Components of the dog's velocity

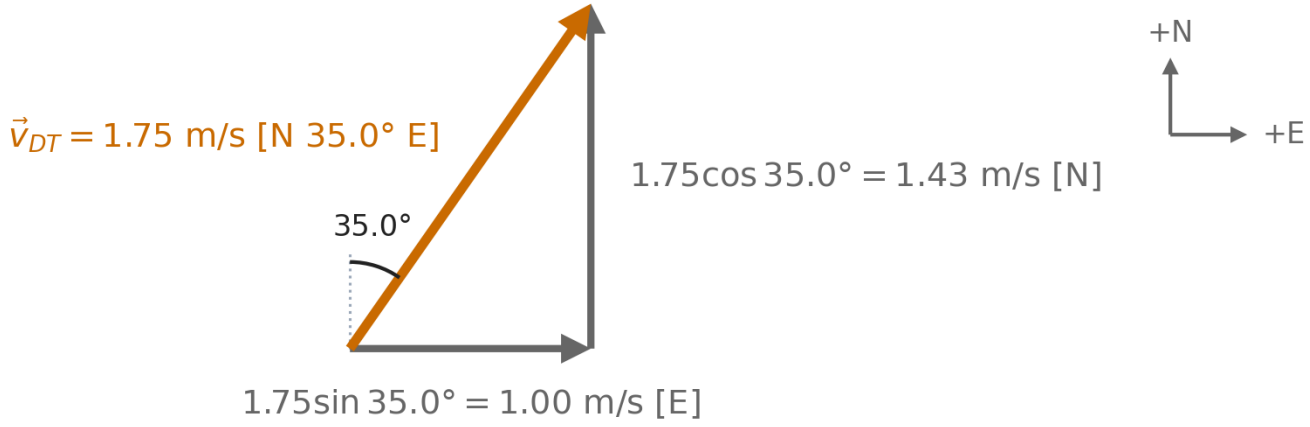
The dog's velocity is on a slant, so we split it into a north part and an east part. That way it can be combined with the truck's purely northward velocity. The 35.0° is measured from north, so the north part is next to the angle (cosine) and the east part is opposite it (sine). v_E and v_N are the east and north parts of the dog's velocity relative to the truck, $\vec{v}_{DT}$.

Formulas: $\text{opposite side} = v\sin\theta$ $\text{adjacent side} = v\cos\theta$

$$v_E = 1.75\sin 35.0° = 1.75(0.5736) = 1.004 \text{ m/s}$$

$$v_N = 1.75\cos 35.0° = 1.75(0.8192) = 1.434 \text{ m/s}$$

Diagram: components of $\vec{v}_{DT}$ (enlarged)

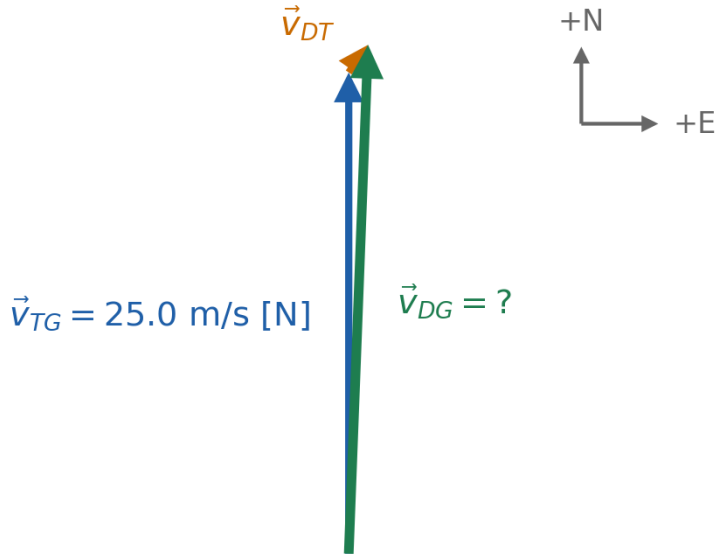


Step 2 – Add the truck's velocity

Dog → truck → ground. The truck moves only north, so it adds to the dog's north part. The east part is unchanged.

north: $25.0 + 1.434 = 26.43$ m/s east: 1.004 m/s

Diagram: $\vec{v}_{TG}$ + $\vec{v}_{DT}$, tip to tail (to scale)



Step 3 – Size and direction

Put the two totals back together with Pythagoras, and find the angle from north with tangent (opposite 1.004 east, adjacent 26.43 north). The result is almost exactly north, because the truck is so much faster than the dog.

Formula (Pythagoras): $c = \sqrt{a^2 + b^2}$

$|\vec{v}_{DG}| = \sqrt{26.43^2 + 1.004^2} = \sqrt{698.7 + 1.0}$

$|\vec{v}_{DG}| = \sqrt{699.7} = 26.45$ m/s

Formula: $\theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right)$

$\theta = \tan^{-1}\left(\frac{1.004}{26.43}\right) = 2.17°$ east of north

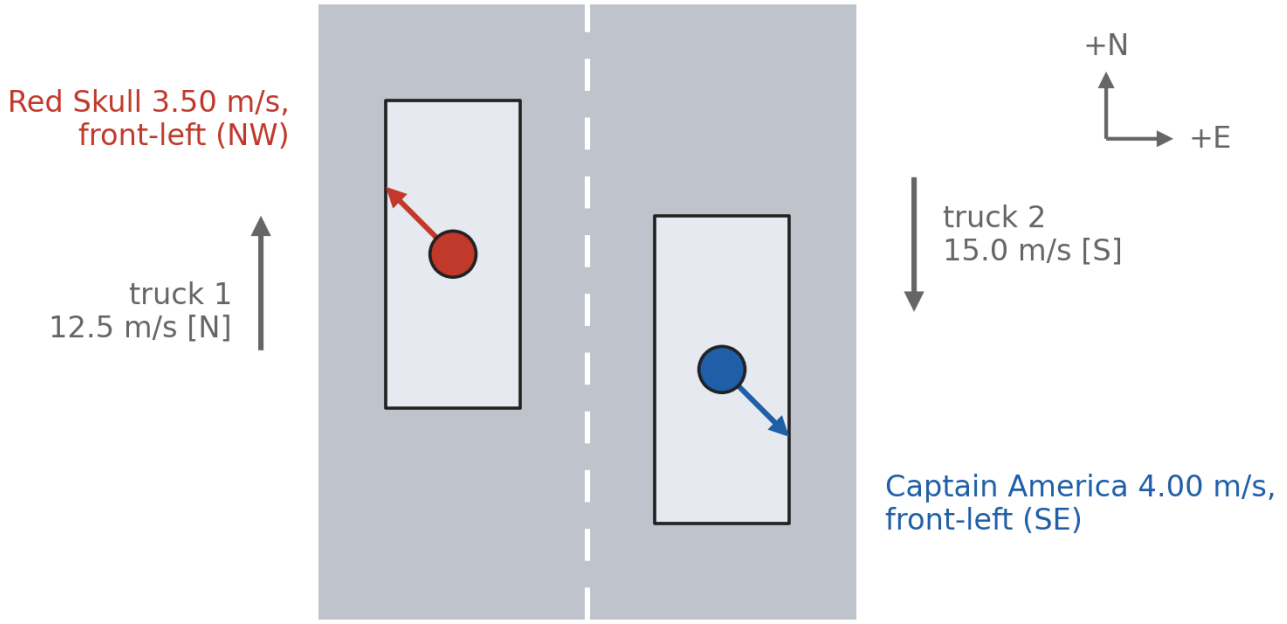
$\vec{v}_{DG}$ = 26.5 m/s [N 2.17° E]

RELATIVE 9 (HONOURS)

Red Skull relative to Captain America

Captain America is chasing Red Skull. He plans to throw his shield to knock down Red Skull but needs to know how fast Red Skull is moving relative to himself. If Red Skull is running at 3.50 m/s at an angle of 45.0° toward the left front of a truck that is moving at 12.5 m/s N, and Captain America is running at 4.00 m/s at an angle of 45.0° to the front left of a second truck moving South at 15.0 m/s, find the relative velocity for Captain America to hit Red Skull.

Picture the situation



Seen from above: two trucks pass each other going opposite ways. Each character runs diagonally toward the front-left of their own truck.

GIVEN

$\vec{v}_{T1,G} = 12.5$ m/s [N]

$\vec{v}_{R,T1} = 3.50$ m/s [NW]

$\vec{v}_{T2,G} = 15.0$ m/s [S]

$\vec{v}_{C,T2} = 4.00$ m/s [SE]

$\vec{v}_{RC} = ?$

Sign convention: R = Red Skull, C = Captain America, T1 = Red Skull's truck, T2 = Captain America's truck, G = ground. A comma separates the two labels when one has a number, so $\vec{v}_{R,T1}$ is Red Skull relative to truck 1. "Left front" is judged facing the way each truck moves: facing north, front-left is north-west; facing south, front-left is south-east.

Step 1 – Red Skull relative to the ground

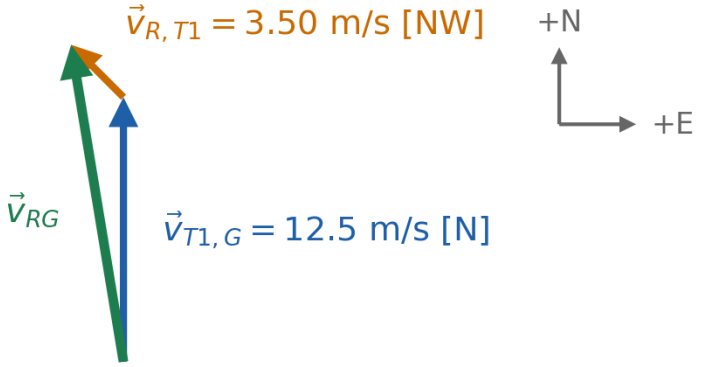
As in Relative 6, first find each person's velocity relative to the ground. Red Skull runs NW on a truck going north. His running velocity is at 45°, so it splits into two equal parts: 3.50 cos 45° = 2.47 m/s west and 2.47 m/s north. The truck adds 12.5 m/s to the north part.

$$\vec{v}_{RG} = \vec{v}_{R,T1} + \vec{v}_{T1,G}$$

Each 45° part: $3.50\cos 45° = 3.50(0.7071) = 2.47$ m/s

west: 2.47 m/s north: $12.5 + 2.47 = 14.97$ m/s

Diagram: Red Skull's ground velocity



Step 2 – Captain America relative to the ground

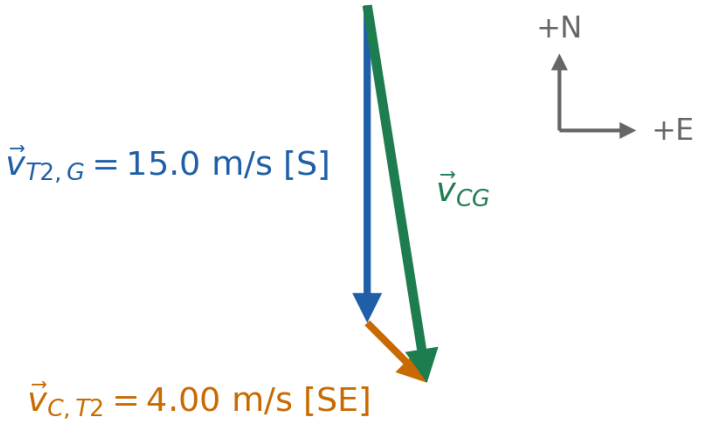
Captain America runs SE on a truck going south. His running splits into 4.00 cos 45° = 2.83 m/s east and 2.83 m/s south. The truck adds 15.0 m/s to the south part.

$$\vec{v}_{CG} = \vec{v}_{C,T2} + \vec{v}_{T2,G}$$

Each 45° part: $4.00\cos 45° = 4.00(0.7071) = 2.83$ m/s

east: 2.83 m/s south: $15.0 + 2.83 = 17.83$ m/s

Diagram: Captain America's ground velocity



Step 3 – Red Skull relative to Captain America

Now look from Captain America's point of view: $\vec{v}_{RC} = \vec{v}_{RG} + \vec{v}_{GC}$, where $\vec{v}_{GC} = -\vec{v}_{CG}$. Reversing Captain America's velocity turns 2.83 east into 2.83 west and 17.83 south into 17.83 north. Add the west parts together and the north parts together, then combine them with Pythagoras and tangent.

$$\vec{v}_{RC} = \vec{v}_{RG} + \vec{v}_{GC} = \vec{v}_{RG} + (-\vec{v}_{CG})$$

$$\text{west: } 2.47 + 2.83 = 5.30 \text{ m/s}$$

$$\text{north: } 14.97 + 17.83 = 32.80 \text{ m/s}$$

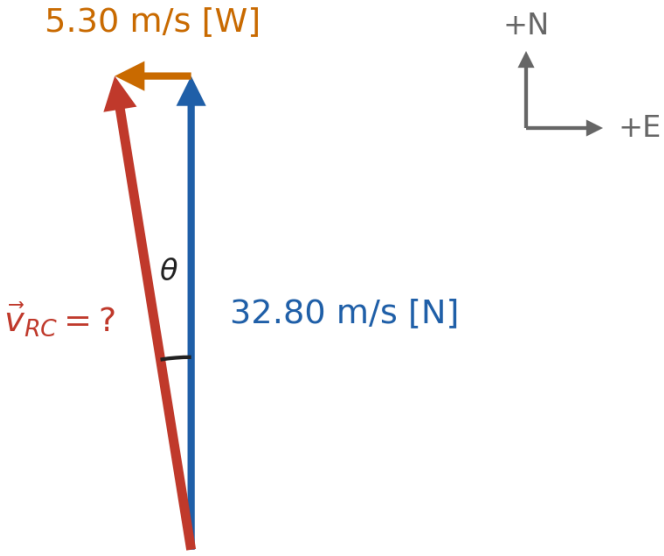
$$\text{Formula (Pythagoras): } c = \sqrt{a^2 + b^2}$$

$$|\vec{v}_{RC}| = \sqrt{5.30^2 + 32.80^2} = \sqrt{28.1 + 1075.8} = \sqrt{1103.9} = 33.2 \text{ m/s}$$

$$\text{Formula: } \theta = \tan^{-1}\left(\frac{\text{opposite}}{\text{adjacent}}\right)$$

$$\theta = \tan^{-1}\left(\frac{5.30}{32.80}\right) = 9.18° \text{ west of north}$$

Diagram: the totals and the resultant



$\vec{v}_{RC}$ = 33.2 m/s [N 9.18° W] (Red Skull relative to Captain America)

Check: the two trucks move in opposite directions, so the relative speed should be close to 12.5 + 15.0 = 27.5 m/s plus the runners' parts along the road (2.47 + 2.83 ≈ 5.3 m/s), with a small west part: 33 m/s fits.